MILITARY STANDARD

RAILCAR LOADING OF HAZARDOUS MATERIALS

MIL-STD-1325-150 (NAVY)

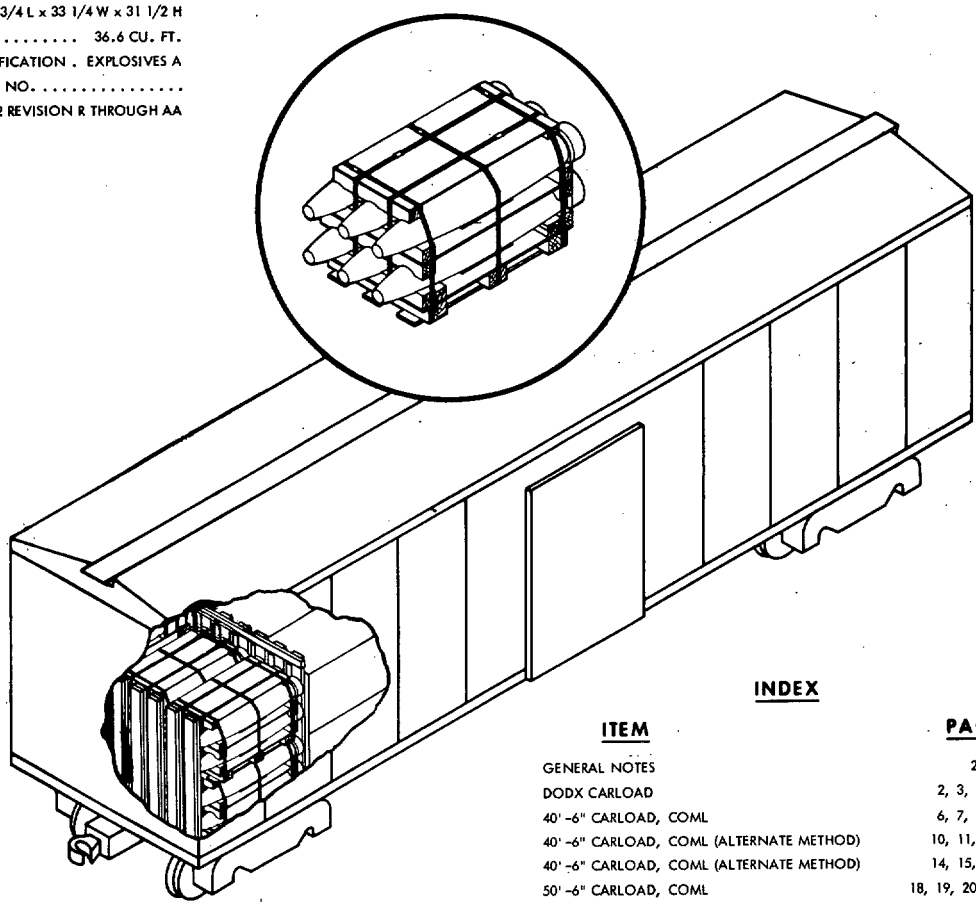
BOMB, GENERAL PURPOSE, MK 82 & MODS (500 LB)
AIR FORCE UNIT LOAD
TPO 1325-294-4152 REVISION R THROUGH A-A

7 SEPTEMBER 1973

SUPERSEDING
WR-52/150 A
10 OCTOBER 1972

UNIT LOAD DATA

WEIGHT ..... 3066 EST.
DIMENSIONS .. 61 3/4 L x 33 1/4 W x 31 1/2 H
CUBE ..... 36.6 CU. FT.
DOT HAZARD CLASSIFICATION . EXPLOSIVES A
AIR FORCE DRAWING NO. .....
.. TPO 1325-294-4152 REVISION R THROUGH AA



NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY).

INDEX

ITEM	PAGE
GENERAL NOTES	2
DODX CARLOAD	2, 3, 4 & 5
40'-6" CARLOAD, COML	6, 7, 8 & 9
40'-6" CARLOAD, COML (ALTERNATE METHOD)	10, 11, 12 & 13
40'-6" CARLOAD, COML (ALTERNATE METHOD)	14, 15, 16 & 17
50'-6" CARLOAD, COML	18, 19, 20, 21 & 22
50'-6" CARLOAD, COML (ALTERNATE METHOD)	24, 25, 26, 27 & 28
50'-6" CARLOAD, COML (ALTERNATE METHOD)	30, 31, 32, 33 & 34
50'-6" CARLOAD, COML (USING LOAD DIVIDERS)	36, 37, 38, 39 & 40
DUMMY UNIT LOAD	42 & 43

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE.

M.S. Gray NWHL 7/24/73
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J.E. Kell 7/26/73
SIGNATURE (AIR) (ORD) SYSCOM, BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

A.F. Grazewski MILITARY ASSISTANT 8/21/73
SIGNATURE DATE

ORIGINATOR Charles McBride 8/24/73
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS."
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 28000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), COMMERCIAL BOXCARS WITH 40 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED, NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF 500 LB GENERAL PURPOSE BOMBS MK 82 AND MODS WITH PLASTIC NOSE PLUGS PALLETIZED IN ACCORDANCE WITH AIR FORCE DRAWING NO. TPO 1325-294-4152 REVISION R THROUGH AA.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. APPLICABLE MATERIAL SPECIFICATIONS:
   DUNNAGE LUMBER - FED. SPEC. MM-L-751
   NAILS - FED. SPEC. FF-N-105
10. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL 1).

50 FT 6 IN. BOXCAR, DODX

1. THE CARLOAD CONSISTS OF 30 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH (SEE MIL-STD-1325 (NAVY). EACH CROSS MEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2200 LBS OF THE LADING.

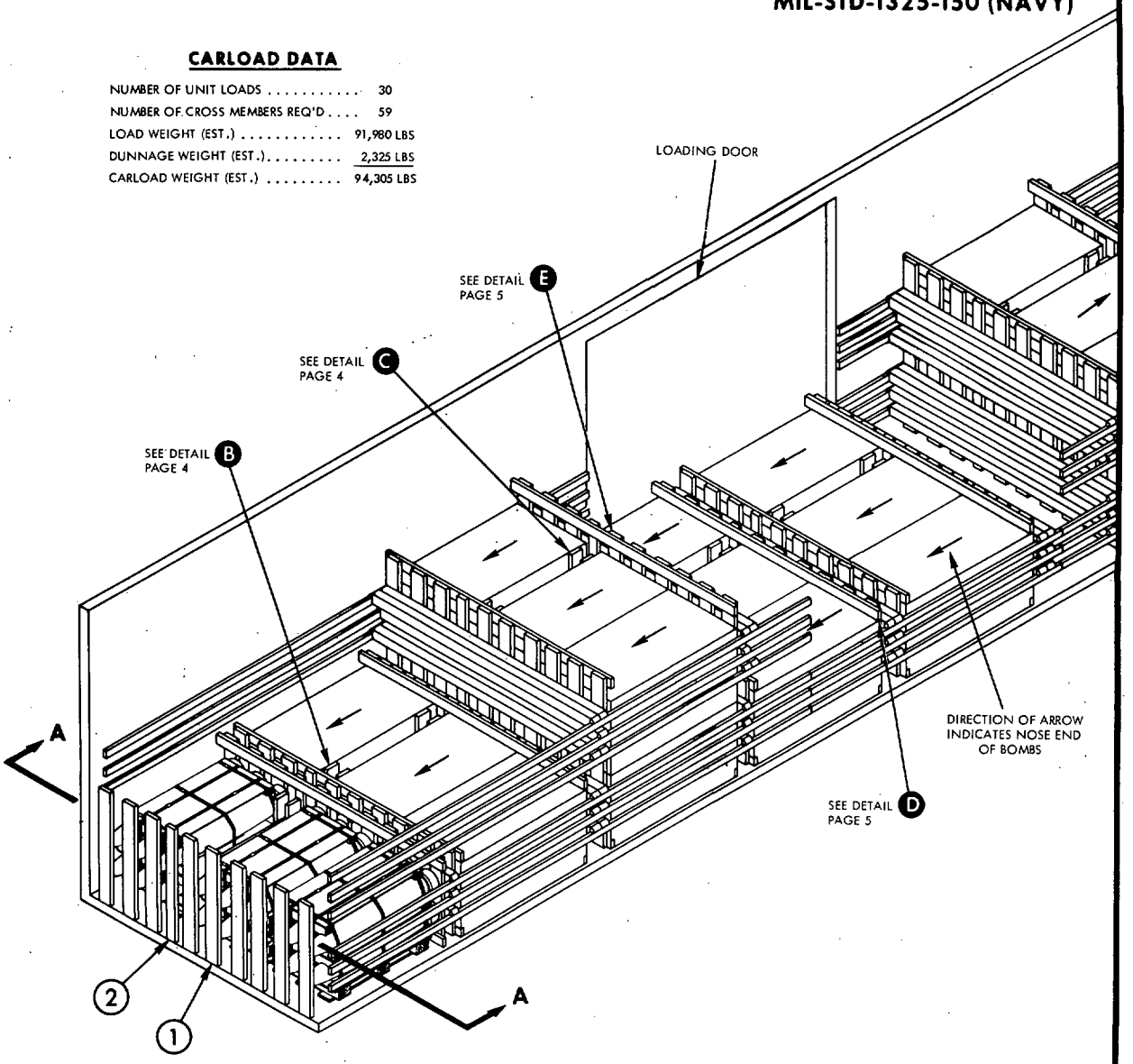
*CROSS BRACE WIDTH SHALL BE A FULL 4 INCHES OR SUFFICIENT WIDTH TO ENSURE NO MORE THAN 1 INCH LATERAL VOID.

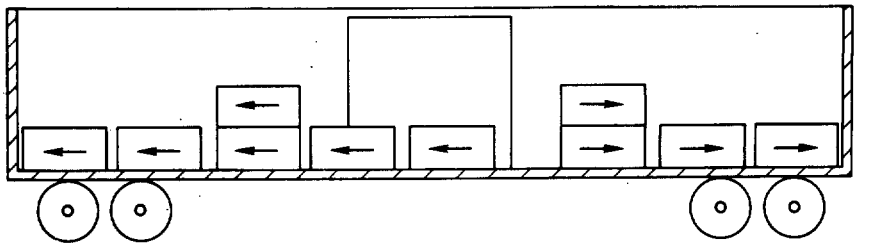
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
LIST OF MATERIALS AND NAILING DATA						
12	SWAY BRACE STOP	2 x 4 x 72	4	SEE 11	-	-
11	SEPARATOR GATE HORIZONTAL	2 x 4 x 107	8	10, 12	3 PER JOINT	10d
10	SEPARATOR GATE VERTICAL	2 x 6 x 72	36	SEE 11	-	-
9	SWAY BRACE STOP	2 x 4 x 36	10	SEE 8	-	-
8	SEPARATOR GATE HORIZONTAL	2 x 4 x 107	20	7, 9	3 PER JOINT	10d
7	SEPARATOR GATE VERTICAL	2 x 6 x 36	90	SEE 8	-	-
6	CROSS BRACE	2 x 4 x 61 1/4*	8	SEE 5	-	-
5	SWAY BRACE VERTICAL	2 x 6 x 60	8	6	3 PER JOINT	16d
4	CROSS BRACE	2 x 4 x 61 1/4*	12	SEE 3	-	-
3	SWAY BRACE VERTICAL	2 x 6 x 30	24	4	3 PER JOINT	16d
2	END WALL MEMBER	2 x 4 x 48	2	CAR WALL	1 PER FOOT	10d
1	END WALL MEMBER	2 x 6 x 48	18	CAR WALL	1 PER FOOT	10d

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CARLOAD DATA

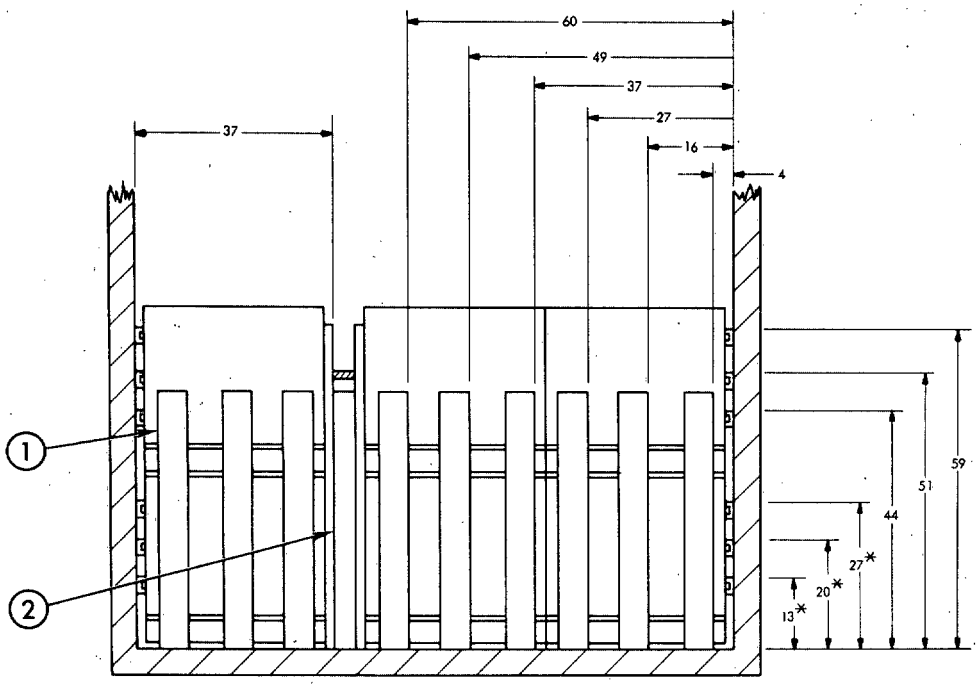
NUMBER OF UNIT LOADS	30
NUMBER OF CROSS MEMBERS REQ'D	59
LOAD WEIGHT (EST.)	91,980 LBS
DUNNAGE WEIGHT (EST.)	2,325 LBS
CARLOAD WEIGHT (EST.)	94,305 LBS





ELEVATION
LOADING PLAN

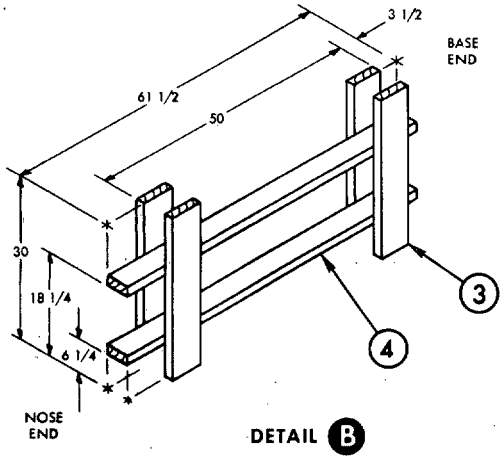
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SECTION A-A

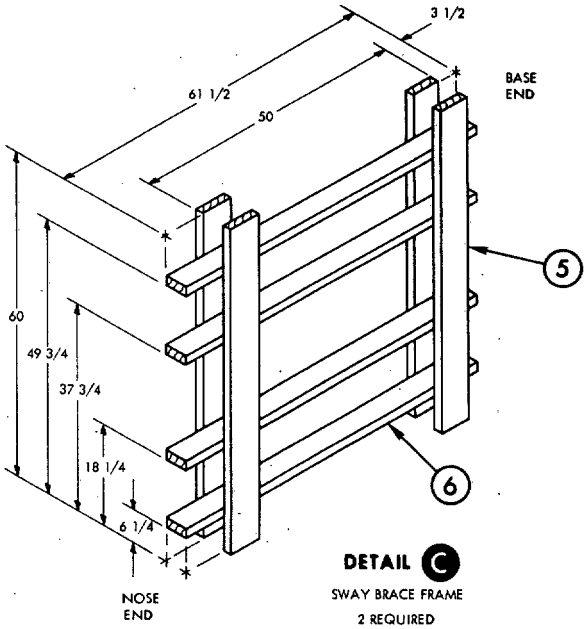
SHOWING LOCATION OF WALL MEMBERS USED
FOR CROSS MEMBERS AND END WALL MEMBERS

*DOORWAY MEMBER LOCATIONS



DETAIL B

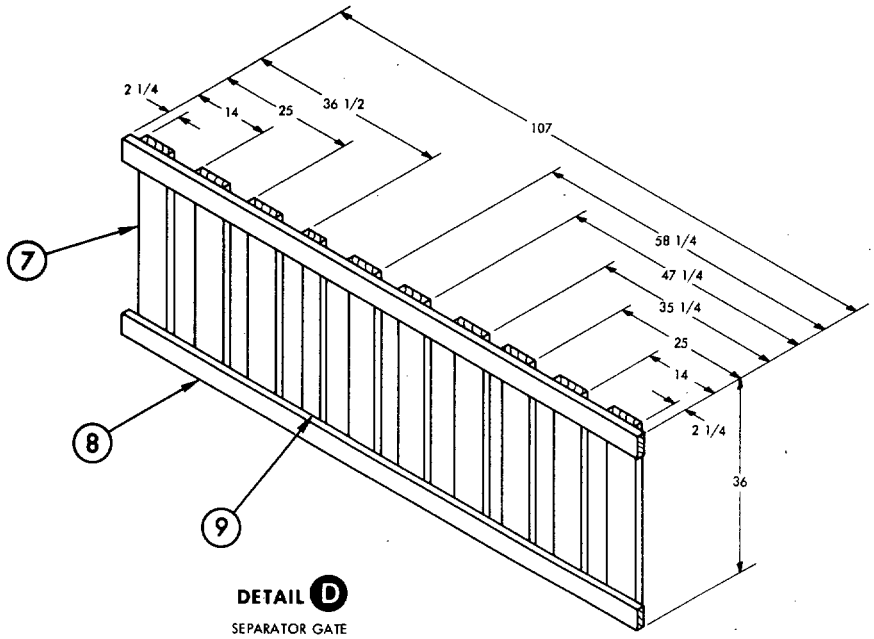
SWAY BRACE FRAME
6 REQUIRED



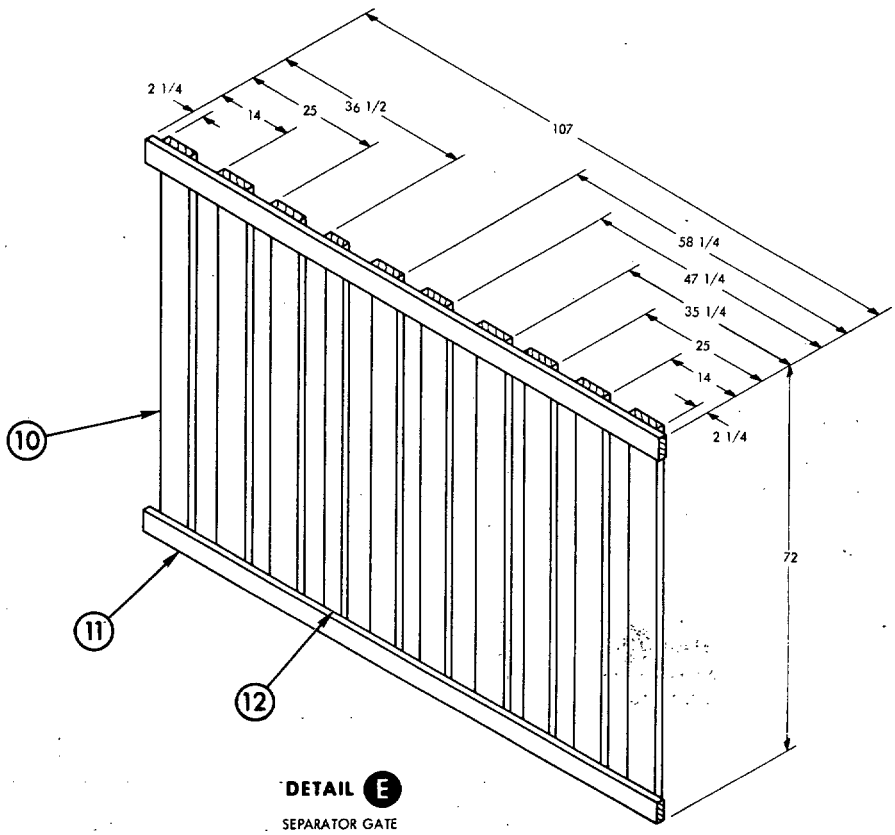
DETAIL C

SWAY BRACE FRAME
2 REQUIRED

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DETAIL D
SEPARATOR GATE
10 REQUIRED

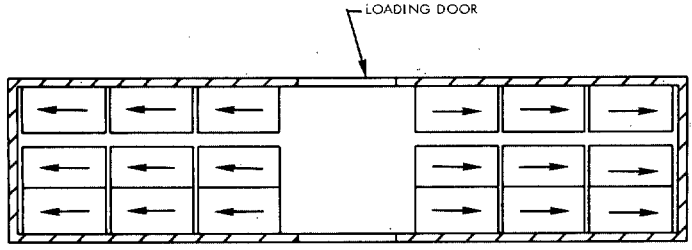


DETAIL E
SEPARATOR GATE
4 REQUIRED

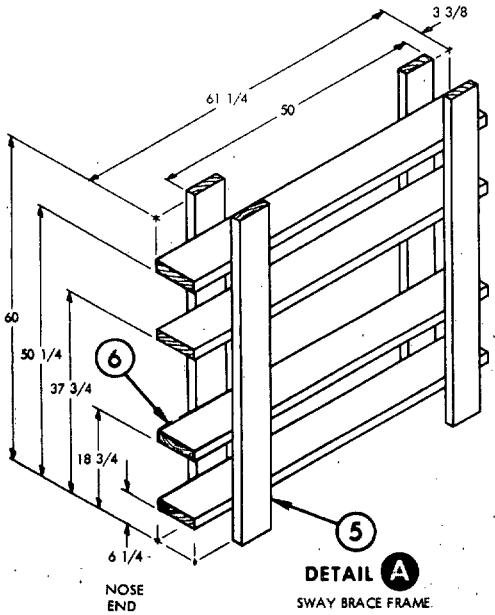
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40 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 36 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED. QUANTITIES CAN ALSO BE ADJUSTED BY SUBSTITUTING DUMMY UNIT LOADS (SEE PAGE 42) AS REQUIRED TO COMPLY WITH LOAD LIMIT OR WEIGHT DISTRIBUTION RULES OF THE A.A.R.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6-FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE, PROVIDED DOORWAY PROTECTION WHEN REQUIRED IS INSTALLED IN ACCORDANCE WITH WR-52.
4. THE SWAY BRACE FRAME SHOWN IN DETAIL A IS FOR BOXCARS WITH 9'-2" OR 9'-4" INSIDE WIDTH. FOR BOXCARS WITH LESS THAN 9'-2" WIDTH, PIECE 6 OF DETAIL A SHOULD BE 2 x 4 MATERIAL AND FOR BOXCARS WIDER THAN 9'-4" PIECE 6 SHOULD BE 2 x 8 MATERIAL.



LOADING PLAN



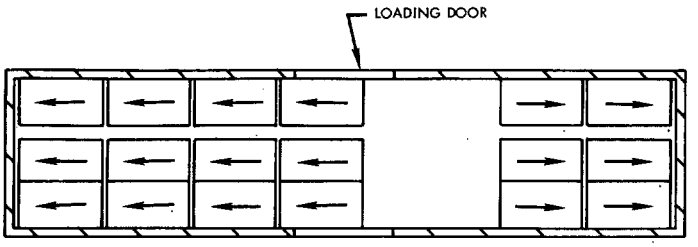
DETAIL A
SWAY BRACE FRAME
6 REQUIRED

*2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE
19	VERTICAL TIE BAR	2 x 4 x 66	6	15	2 PER JOINT	16d
18	HORIZONTAL TIE BAR	2 x 4 x CAR WIDTH-1	4	15	2 PER JOINT	16d
17	HOLD DOWN CLEAT	2 x 6 x 24	4	16	5	10d
16	GATE HOLD DOWN	2 x 6 x 10 FT	2	CAR WALL	5 EACH END	10d
15	STRUT	4 x 4 x WEDGE FIT*	24	11	2 PER JOINT	16d
14	SWAY BRACE STOP	2 x 6 x 30	2	11	3 PER JOINT	10d
13	CENTER GATE HORIZONTAL	2 x 4 x CAR WIDTH-1	2	11	3 PER JOINT	10d
12	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	11	3 PER JOINT	10d
11	CENTER GATE VERTICAL	2 x 6 x 72	12	SEE 10	-	-
10	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	11	3 PER JOINT	10d
9	SWAY BRACE STOP	2 x 4 x 72	4	SEE 8	-	-
8	SEPARATOR GATE HORIZONTAL	2 x 4 x CAR WIDTH-1	8	7,9	3 PER JOINT	10d
7	SEPARATOR GATE VERTICAL	2 x 6 x 72	36	SEE 8	-	-
6	CROSS BRACE	2 x 6 x 61 1/4	24	SEE 5	-	-
5	SWAY BRACE VERTICAL	2 x 6 x 60	24	6	3 PER JOINT	16d
4	END WALL MEMBER	2 x 4 x 72	2	3	1 PER FOOT	10d
3	END WALL MEMBER	2 x 4 x 72	2	CAR WALL	1 PER FOOT	10d
2	END WALL MEMBER	2 x 6 x 72	18	1	1 PER FOOT	10d
1	END WALL MEMBER	2 x 6 x 72	18	CAR WALL	1 PER FOOT	10d

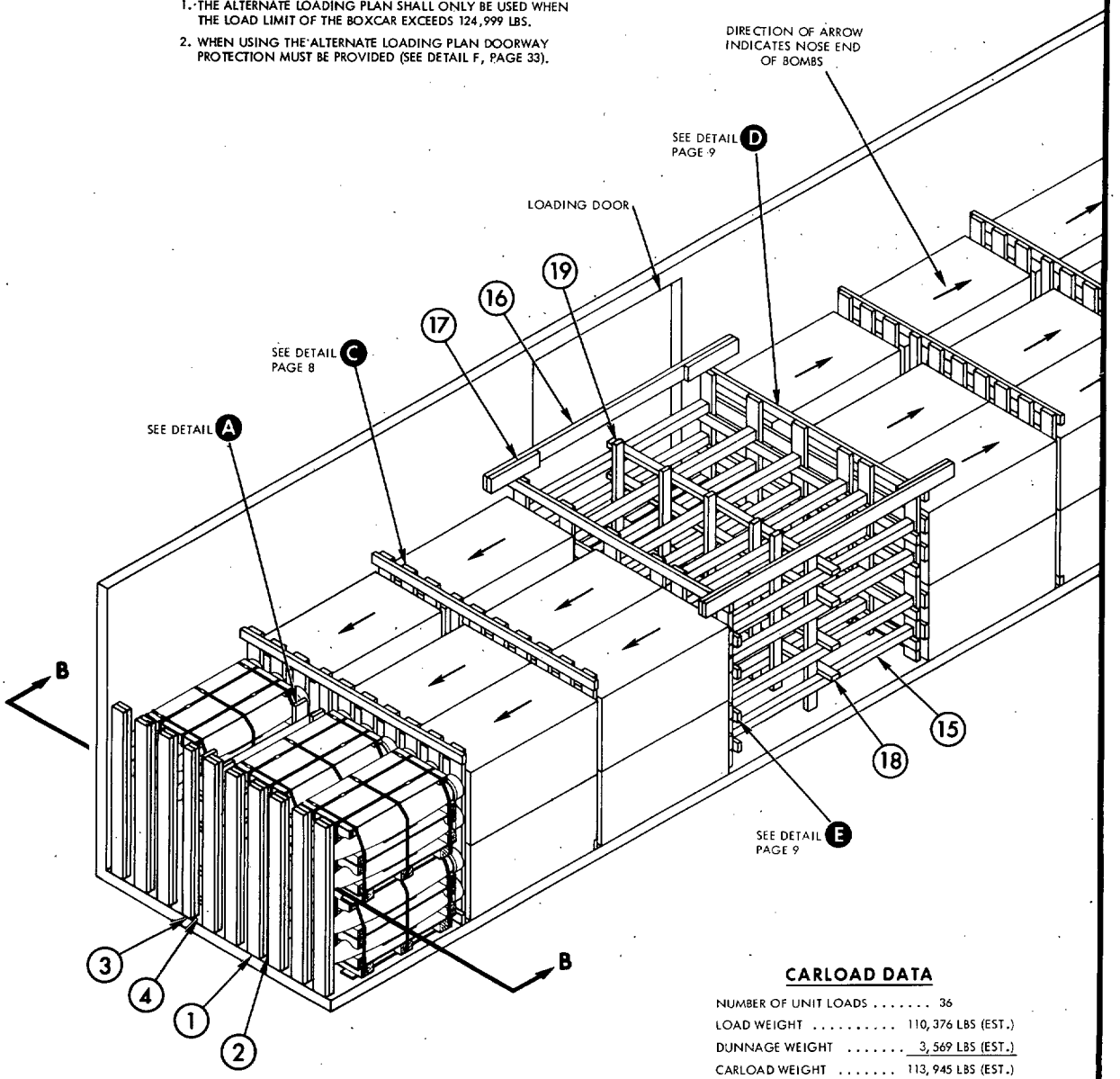
LIST OF MATERIALS AND NAILING DATA

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ALTERNATE LOADING PLAN

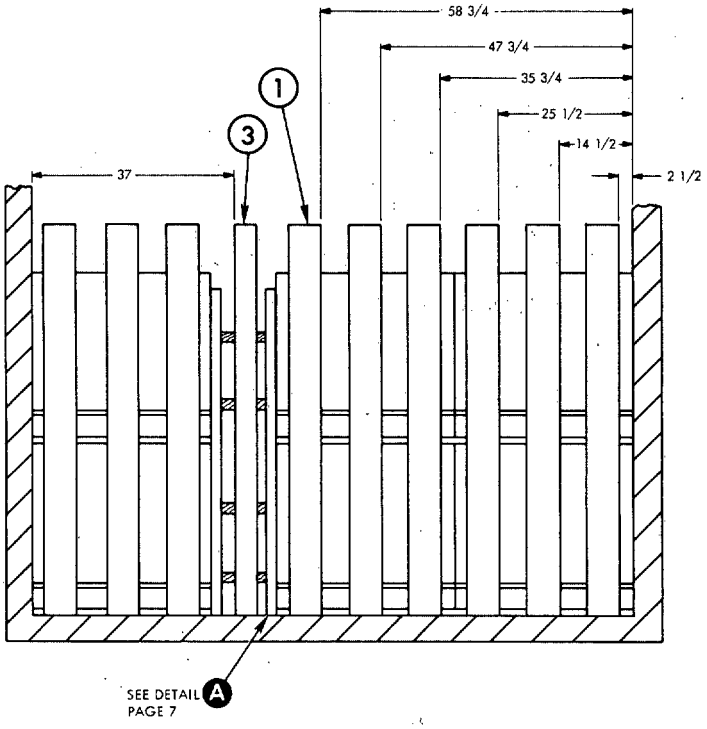
1. THE ALTERNATE LOADING PLAN SHALL ONLY BE USED WHEN THE LOAD LIMIT OF THE BOXCAR EXCEEDS 124,999 LBS.
2. WHEN USING THE ALTERNATE LOADING PLAN DOORWAY PROTECTION MUST BE PROVIDED (SEE DETAIL F, PAGE 33).



CARLOAD DATA

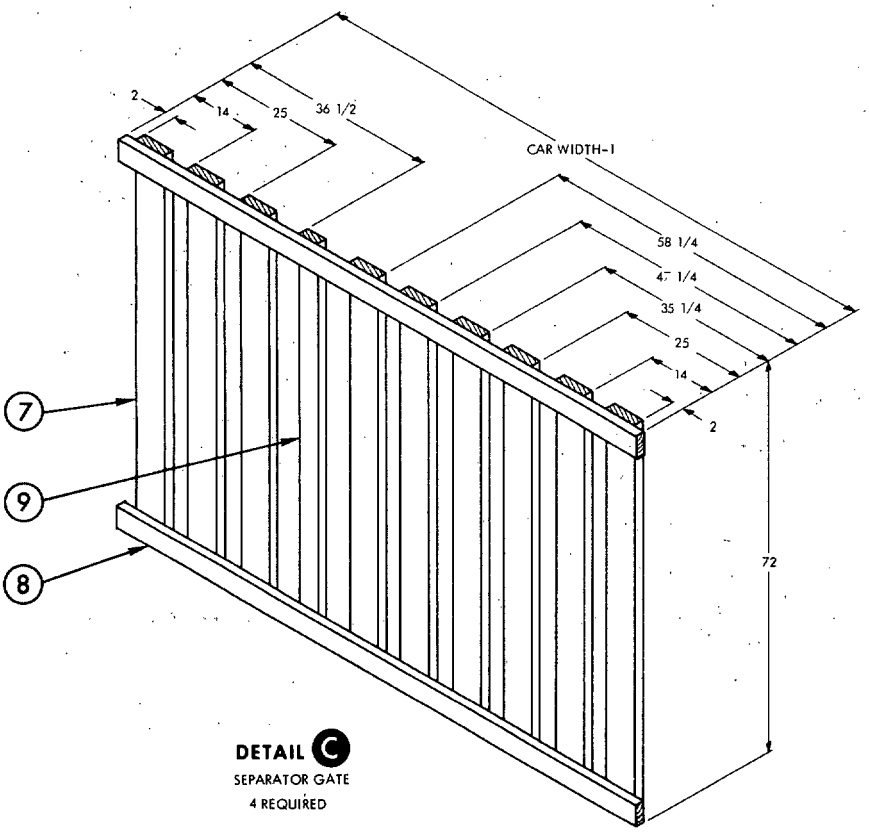
NUMBER OF UNIT LOADS	36
LOAD WEIGHT	110,376 LBS (EST.)
DUNNAGE WEIGHT	3,569 LBS (EST.)
CARLOAD WEIGHT	113,945 LBS (EST.)

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SECTION B-B

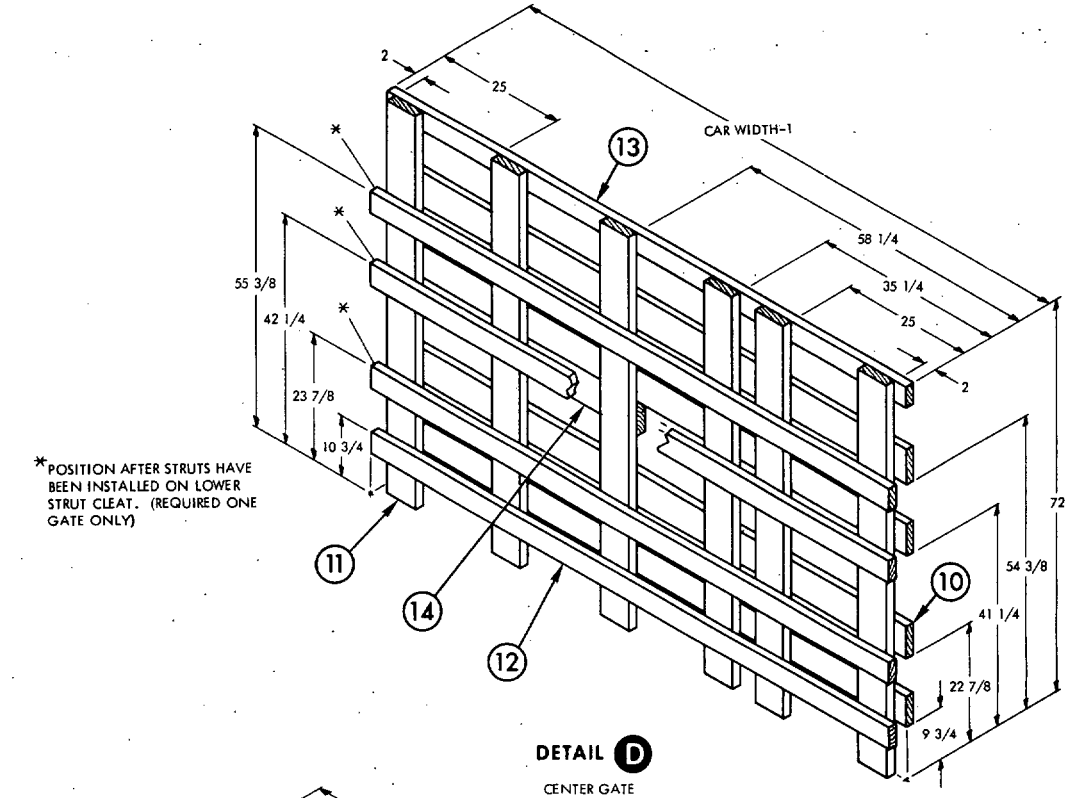
SHOWING LOCATION OF END WALL MEMBERS
(REVERSE DIMENSIONS FOR OTHER END OF CAR)



DETAIL C

SEPARATOR GATE
4 REQUIRED

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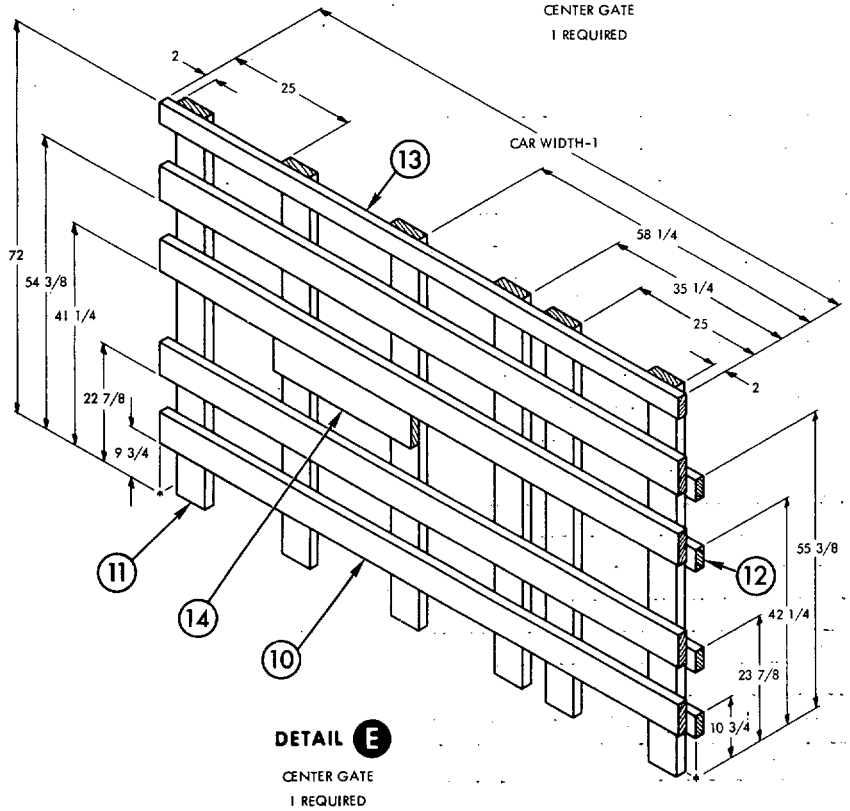


*POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT. (REQUIRED ONE GATE ONLY)

DETAIL D

CENTER GATE

1 REQUIRED



DETAIL E

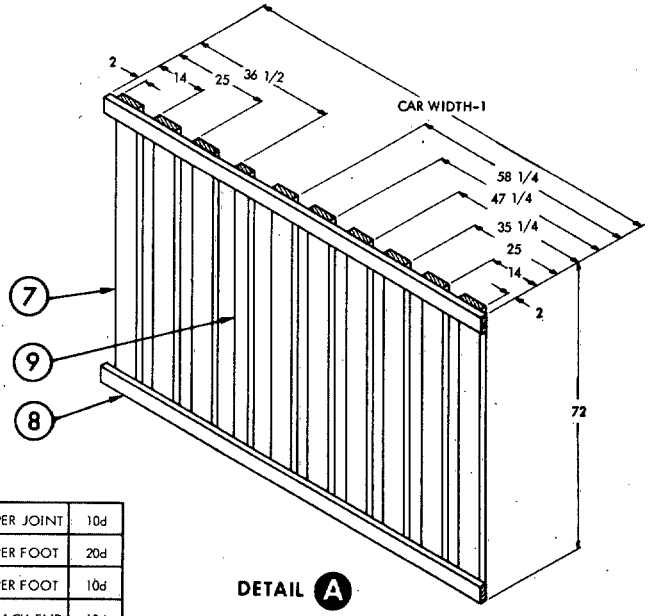
CENTER GATE

1 REQUIRED

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40 FT 6 IN. BOXCAR, COMMERCIAL (ALTERNATE METHOD)

1. THE CARLOAD CONSISTS OF 40 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED. QUANTITIES CAN ALSO BE ADJUSTED BY SUBSTITUTING DUMMY UNIT LOADS (SEE PAGE 42) AS REQUIRED TO COMPLY WITH LOAD LIMIT OR WEIGHT DISTRIBUTION RULES OF THE A.A.R.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE.
4. THE SWAY BRACE FRAME SHOWN IN DETAIL A IS FOR BOXCARS WITH 9'-2" OR 9'-4" INSIDE WIDTH. FOR BOXCARS WITH LESS THAN 9'-2" WIDTH, PIECE 6 OF DETAIL A SHOULD BE 2 x 4 MATERIAL AND FOR BOXCARS WIDER THAN 9'-4" PIECE 6 SHOULD BE 2 x 8 MATERIAL.



DETAIL A

SEPARATOR GATE

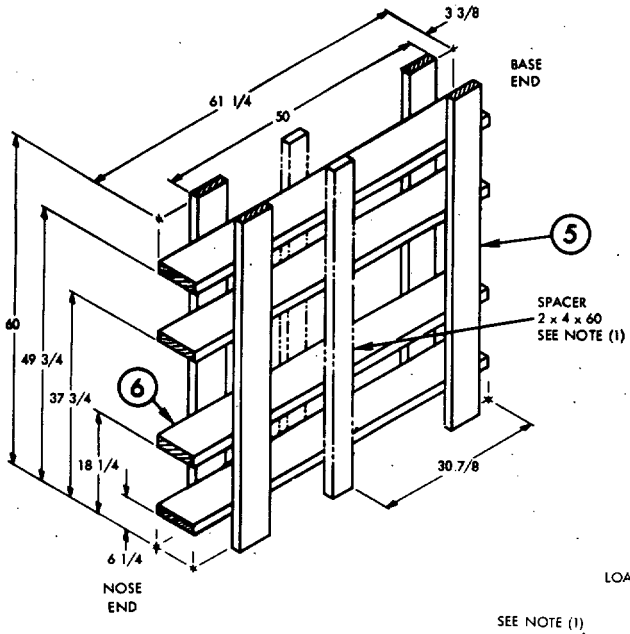
5 REQUIRED

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE NAILS
19	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	8	18	2 PER JOINT	10d
18	VERTICAL DOORWAY MEMBER	2 x 3 x 72	4	DOOR POST	2 PER FOOT	20d
17	HOLD DOWN CLEAT	2 x 6 x 60	2	16	2 PER FOOT	10d
16	GATE HOLD DOWN	2 x 6 x 10 FT	2	CAR WALL	5 EACH END	10d
15	STRUT	4 x 4 x WEDGE FIT*	24	11	2 PER JOINT	16d
14	SWAY BRACE STOP	2 x 6 x 30	2	11	3 PER JOINT	10d
13	CENTER GATE HORIZONTAL	2 x 4 x CAR WIDTH-1	2	11	3 PER JOINT	10d
12	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	11	3 PER JOINT	10d
11	CENTER GATE VERTICAL	2 x 6 x 72	12	SEE 10	-	-
10	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	11	3 PER JOINT	10d
9	SWAY BRACE STOP	2 x 4 x 72	5	SEE 8	-	-
8	SEPARATOR GATE HORIZONTAL	2 x 4 x CAR WIDTH-1	10	7,9	3 PER JOINT	10d
7	SEPARATOR GATE VERTICAL	2 x 6 x 72	45	SEE 8	-	-
6	CROSS BRACE	2 x 6 x 61 1/4	28	SEE 5	-	-
5	SWAY BRACE VERTICAL	2 x 6 x 60	28	6	3 PER JOINT	16d
4	END WALL MEMBER	2 x 4 x 72	2	3	1 PER FOOT	10d
3	END WALL MEMBER	2 x 4 x 72	2	CAR WALL	1 PER FOOT	10d
2	END WALL MEMBER	2 x 6 x 72	18	1	1 PER FOOT	10d
1	END WALL MEMBER	2 x 6 x 72	18	CAR WALL	1 PER FOOT	10d

LIST OF MATERIALS AND NAILING DATA

*2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

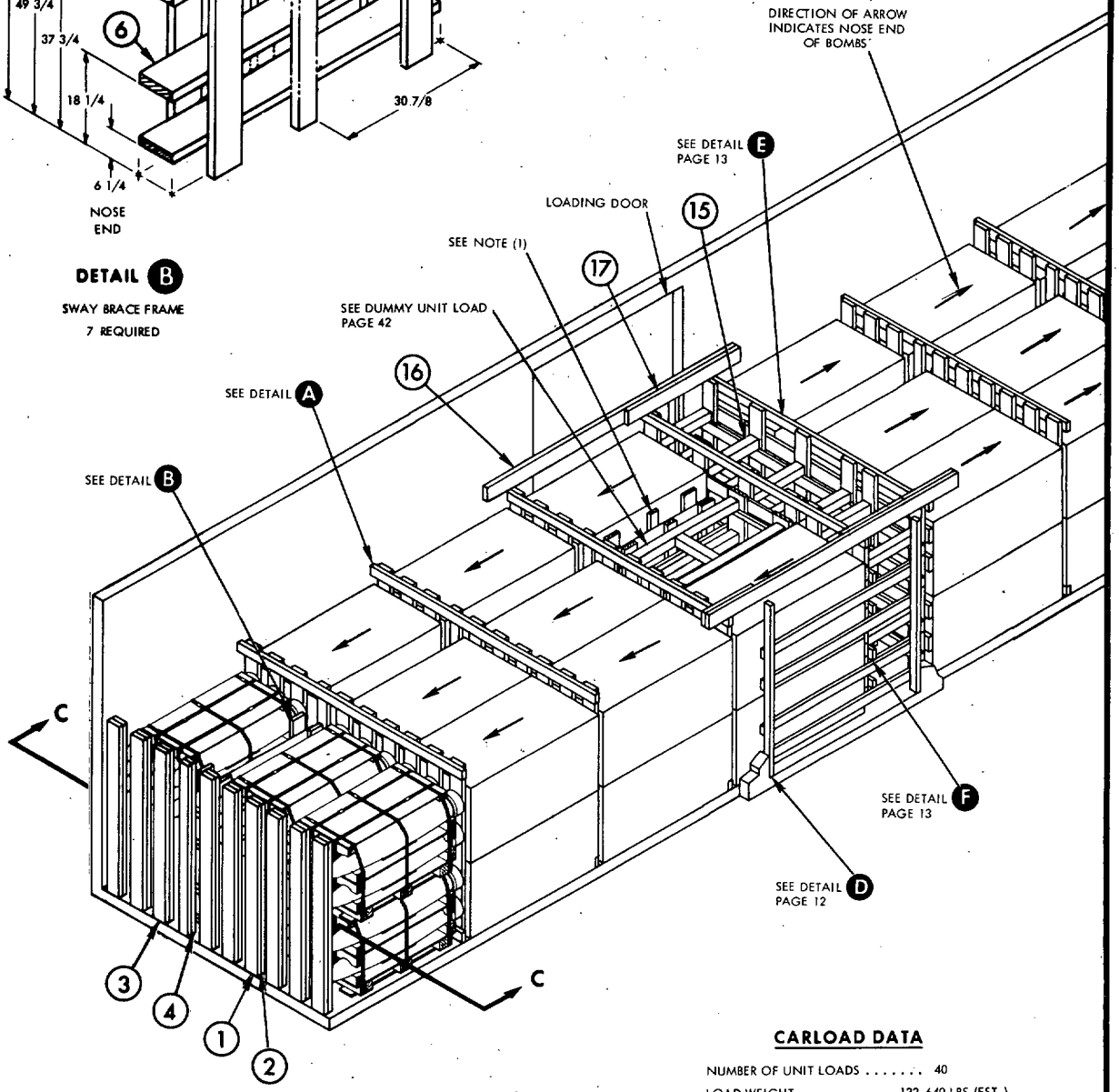
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DETAIL B

SWAY BRACE FRAME
7 REQUIRED

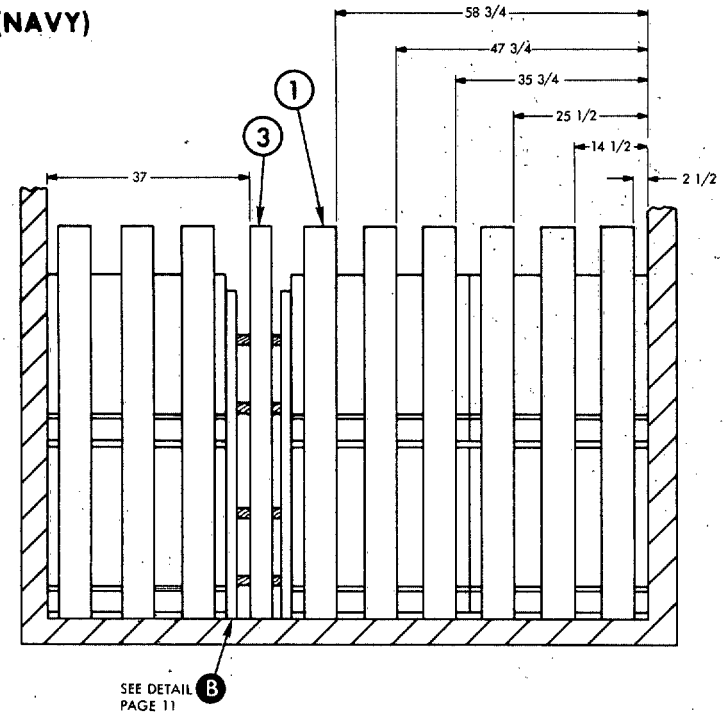
NOTE:
(1) WHEN POSITIONING DUMMY LOAD NEXT TO SWAY BRACE FRAME, 2 x 4 SPACERS MUST BE INSTALLED ON EACH SIDE OF FRAME. THIS IS NOT REQUIRED WHEN 3 DUMMY LOADS ARE LOADED ACROSS THE CAR.



CARLOAD DATA

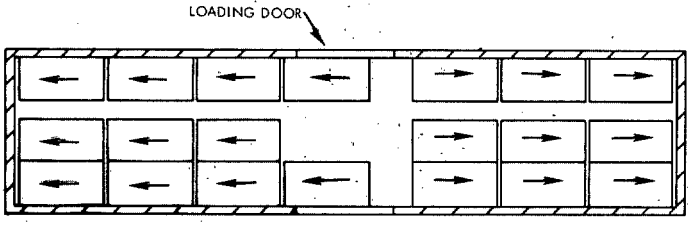
NUMBER OF UNIT LOADS	40
LOAD WEIGHT	122,640 LBS (EST.)
DUNNAGE WEIGHT	3,816 LBS (EST.)
CARLOAD WEIGHT	126,456 LBS (EST.)

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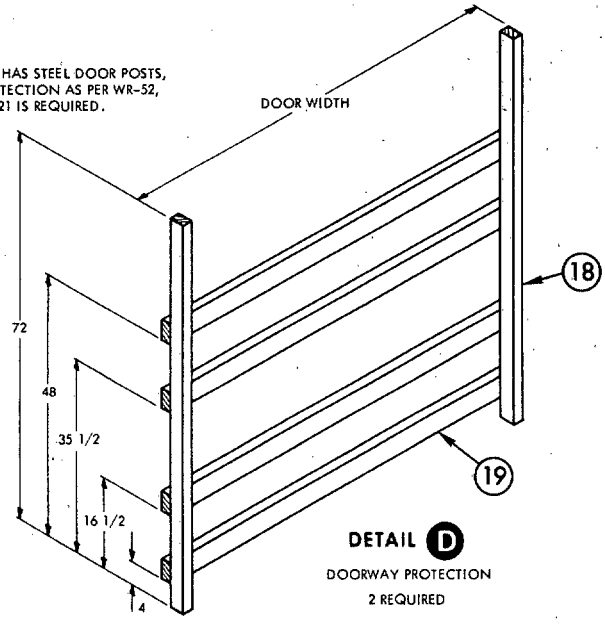
SECTION C-C

SHOWING LOCATION OF END WALL MEMBERS
(REVERSE DIMENSIONS FOR OTHER END OF CAR)



LOADING PLAN

WHEN BOXCAR HAS STEEL DOOR POSTS, DOORWAY PROTECTION AS PER WR-52, PAGE 35, FIG. 21 IS REQUIRED.

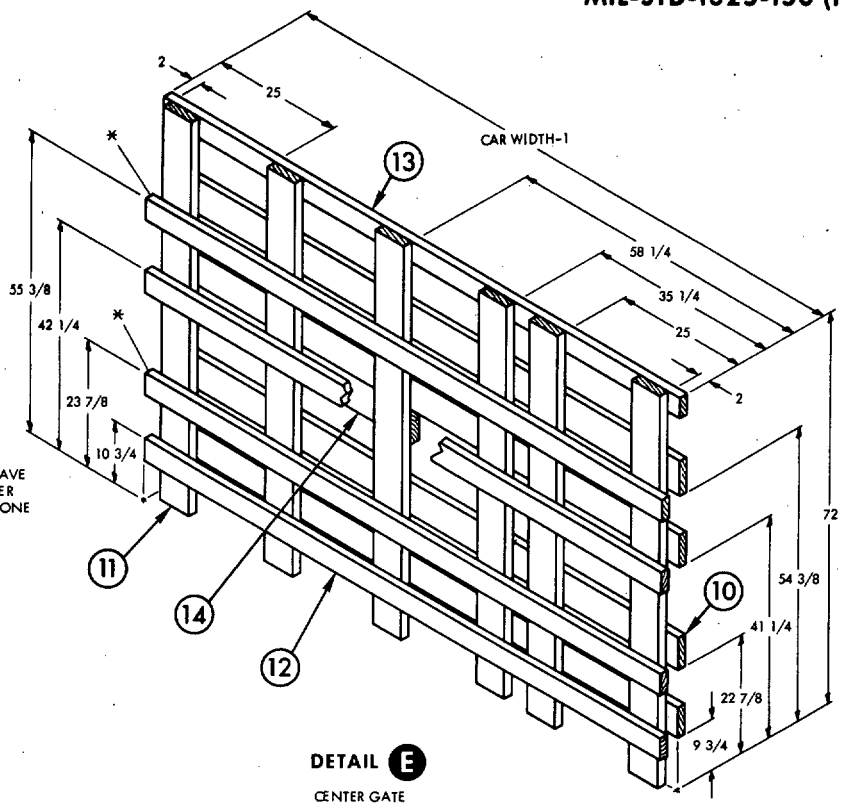


DETAIL D

DOORWAY PROTECTION

2 REQUIRED

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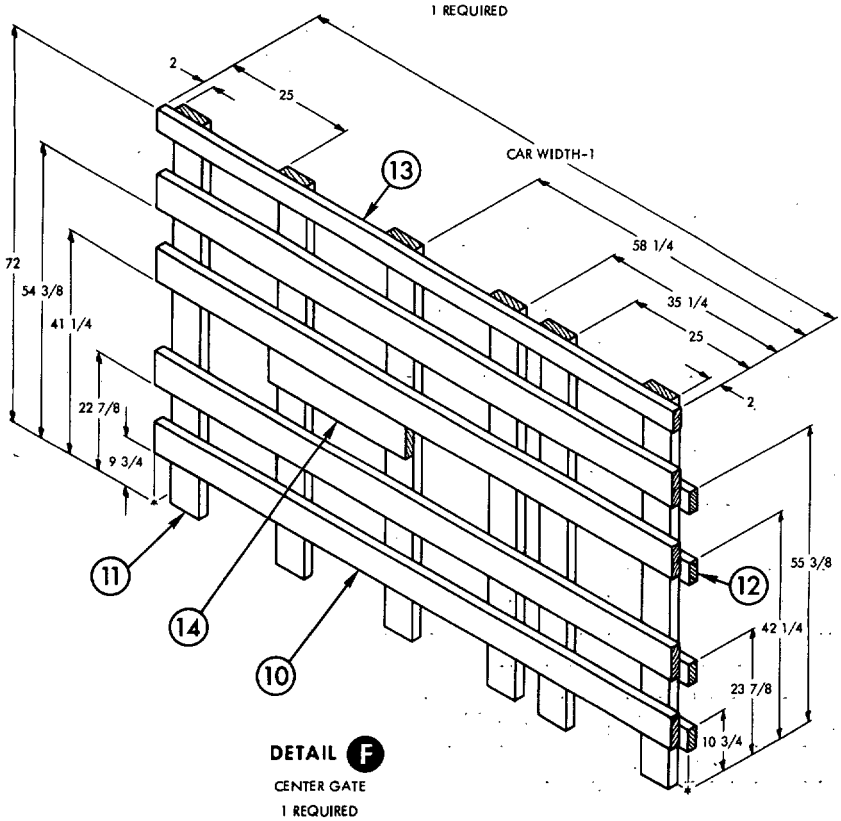


*POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT. (REQUIRED ONE GATE ONLY)

DETAIL E

CENTER GATE

1 REQUIRED



DETAIL F

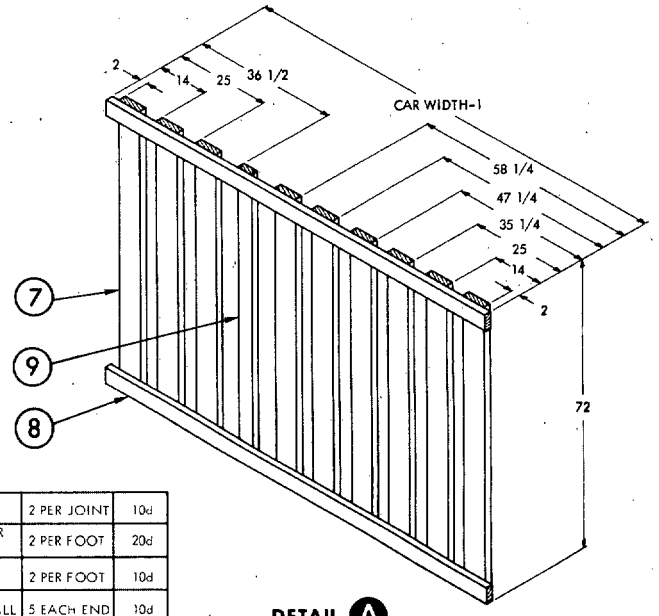
CENTER GATE

1 REQUIRED

Source: https://assist.dla.mil -- Downloaded: 20261O-26T07:45Z
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40 FT 6 IN. BOXCAR, COMMERCIAL (ALTERNATE METHOD)

1. THE CARLOAD CONSISTS OF 39 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED. QUANTITIES CAN ALSO BE ADJUSTED BY SUBSTITUTING DUMMY UNIT LOADS (SEE PAGE 42) AS REQUIRED TO COMPLY WITH LOAD LIMIT OR WEIGHT DISTRIBUTION RULES OF THE A.A.R.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE.
4. THE SWAY BRACE FRAME SHOWN IN DETAIL A IS FOR BOXCARS WITH 9'-2" OR 9'-4" INSIDE WIDTH. FOR BOXCARS WITH LESS THAN 9'-2" WIDTH, PIECE 6 OF DETAIL A SHOULD BE 2 x 4 MATERIAL AND FOR BOXCARS WIDER THAN 9'-4" PIECE 6 SHOULD BE 2 x 8 MATERIAL.



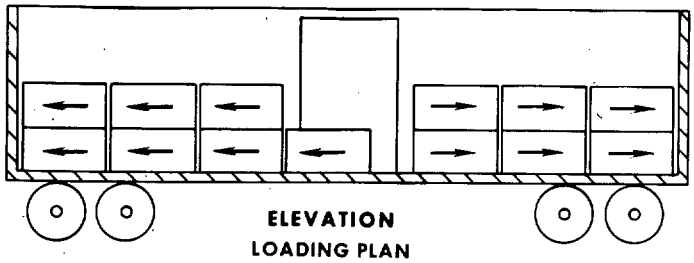
DETAIL A
SEPARATOR GATE
5 REQUIRED

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE
19	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	8	18	2 PER JOINT	10d
18	VERTICAL DOORWAY MEMBER	2 x 3 x 72	4	DOOR POST	2 PER FOOT	20d
17	HOLD DOWN CLEAT	2 x 6 x 60	2	16	2 PER FOOT	10d
16	GATE HOLD DOWN	2 x 6 x 10 FT	2	CAR WALL	5 EACH END	10d
15	STRUT	4 x 4 x WEDGE FIT *	24	11	2 PER JOINT	16d
14	SWAY BRACE STOP	2 x 6 x 30	2	11	3 PER JOINT	10d
13	CENTER GATE HORIZONTAL	2 x 4 x CAR WIDTH-1	2	11	3 PER JOINT	10d
12	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	11	3 PER JOINT	10d
11	CENTER GATE VERTICAL	2 x 6 x 72	12	SEE 10	-	-
10	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	11	3 PER JOINT	10d
9	SWAY BRACE STOP	2 x 4 x 72	5	SEE 8	-	-
8	SEPARATOR GATE HORIZONTAL	2 x 4 x CAR WIDTH-1	10	7,9	3 PER JOINT	10d
7	SEPARATOR GATE VERTICAL	2 x 6 x 72	45	SEE 8	-	-
6	CROSS BRACE	2 x 6 x 61 1/4	28	SEE 5	-	-
5	SWAY BRACE VERTICAL	2 x 6 x 60	28	6	3 PER JOINT	16d
4	END WALL MEMBER	2 x 4 x 72	2	3	1 PER FOOT	10d
3	END WALL MEMBER	2 x 4 x 72	2	CAR WALL	1 PER FOOT	10d
2	END WALL MEMBER	2 x 6 x 72	18	1	1 PER FOOT	10d
1	END WALL MEMBER	2 x 6 x 72	18	CAR WALL	1 PER FOOT	10d

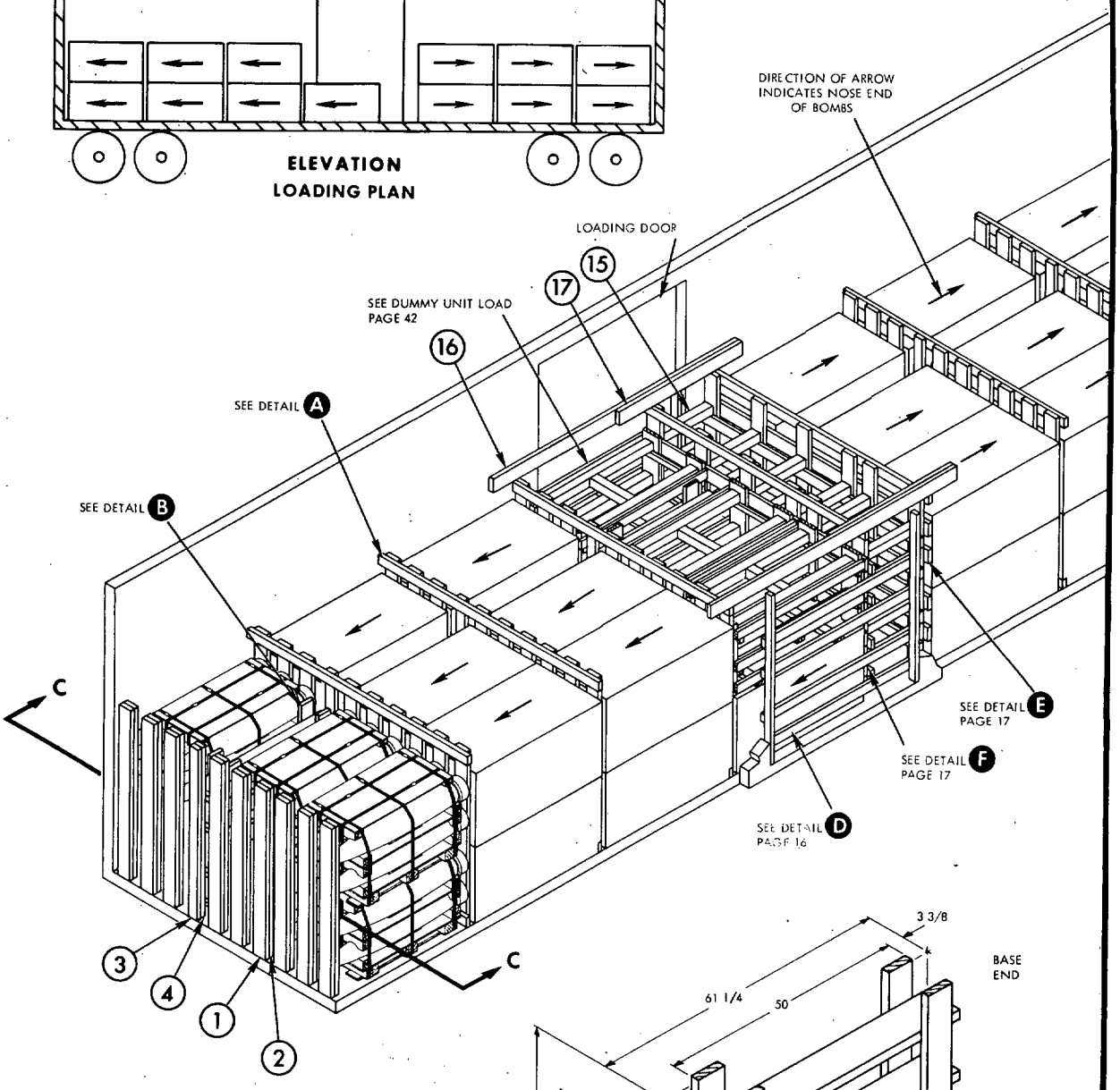
LIST OF MATERIALS AND NAILING DATA

*2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

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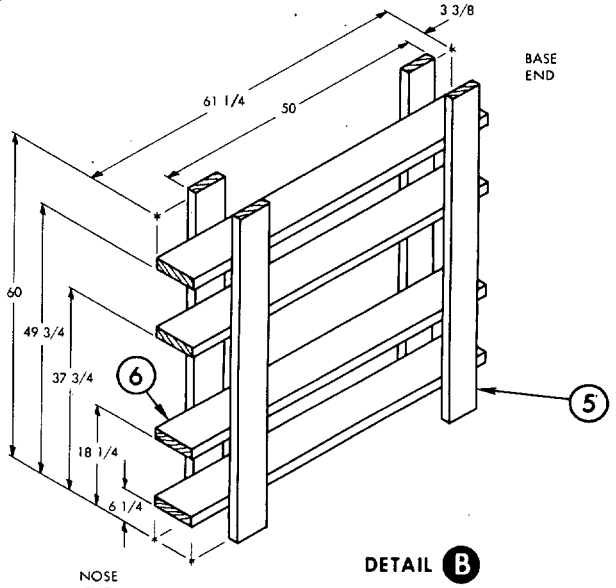


ELEVATION
LOADING PLAN



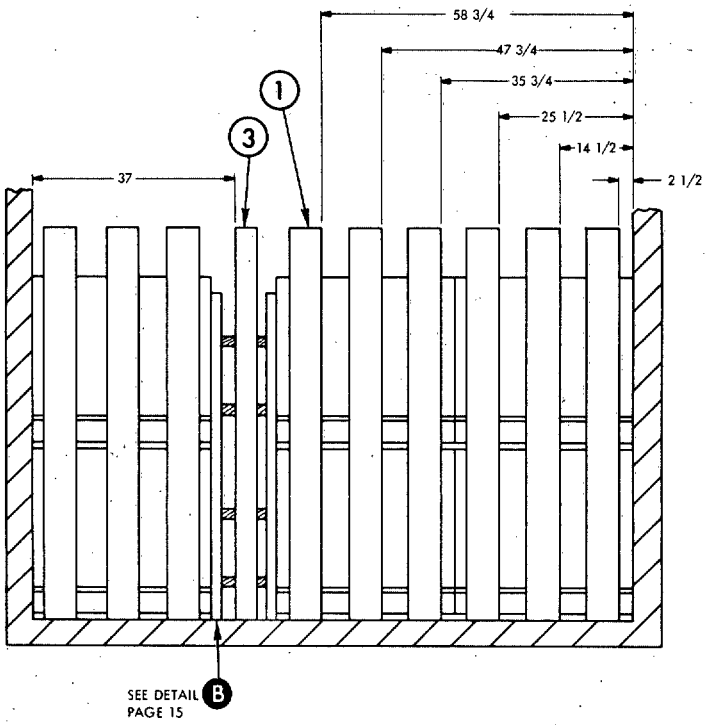
CARLOAD DATA

NUMBER OF UNIT LOADS	39
LOAD WEIGHT	119,574 LBS (EST.)
DUNNAGE WEIGHT	4,022 LBS (EST.)
CARLOAD WEIGHT	123,596 LBS (EST.)



DETAIL B
SWAY BRACE FRAME
7 REQUIRED

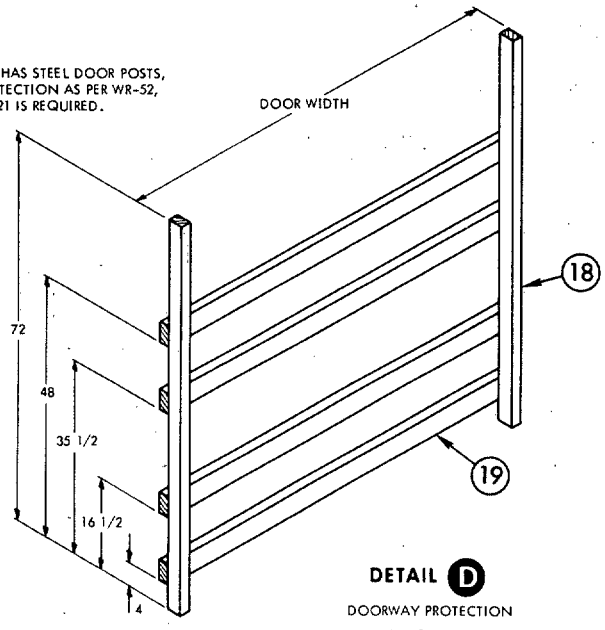
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SECTION C-C

SHOWING LOCATION OF END WALL MEMBERS
(REVERSE DIMENSIONS FOR OTHER END OF CAR)

WHEN BOXCAR HAS STEEL DOOR POSTS, DOORWAY PROTECTION AS PER WR-52, PAGE 35, FIG. 21 IS REQUIRED.

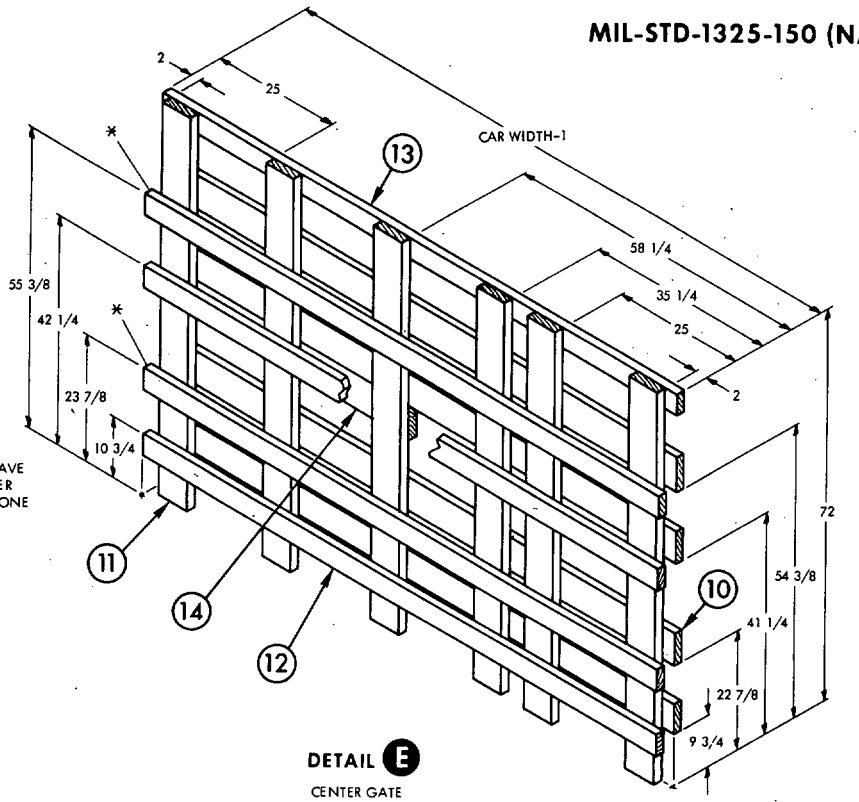


DETAIL D

DOORWAY PROTECTION

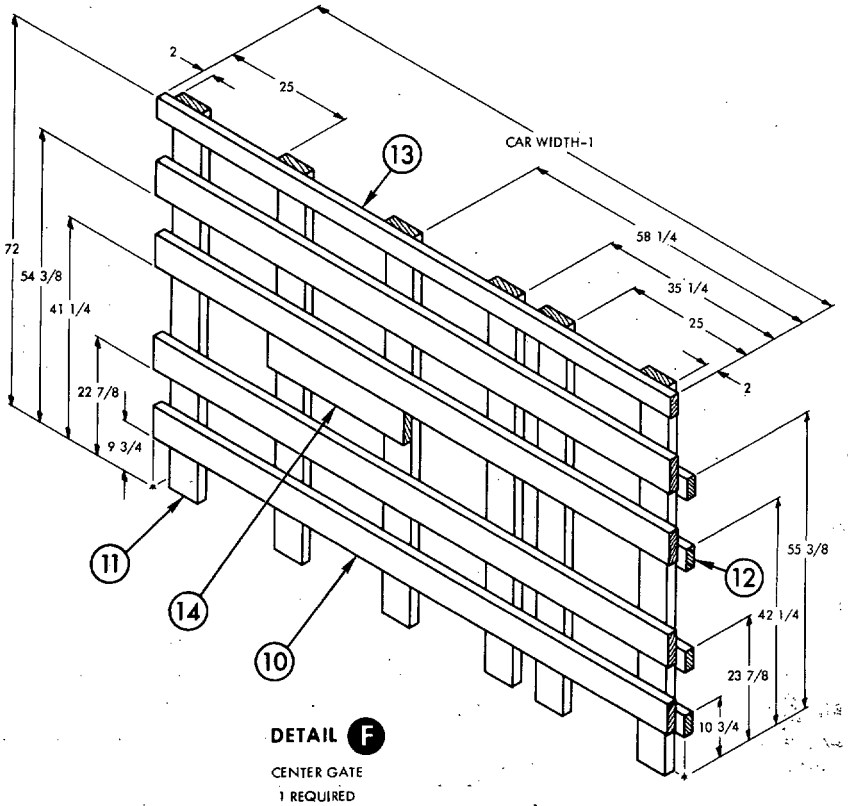
2 REQUIRED

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:45Z
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*POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT. (REQUIRED ONE GATE ONLY)

DETAIL E
CENTER GATE
1 REQUIRED



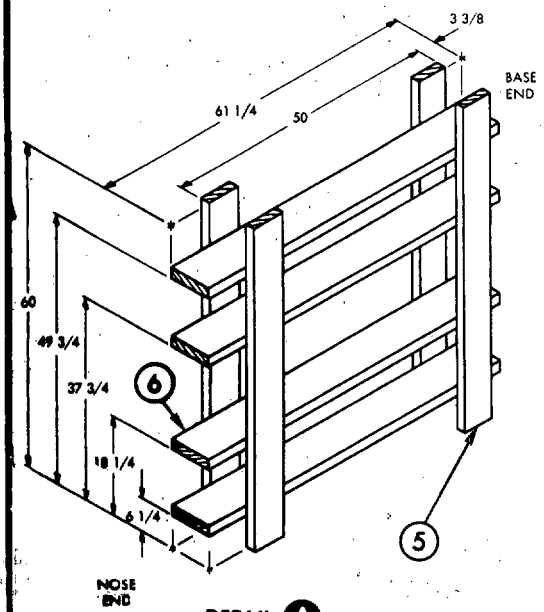
DETAIL F
CENTER GATE
1 REQUIRED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
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50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 48 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING, WHEN LOAD LIMIT OF CAR PERMITS (SEE NOTE 4 OF GENERAL NOTES, PAGE 2) BOXCAR MAY ALSO BE LOADED IN THE 5 AND 3 PATTERN FOR A TOTAL OF 48 UNIT LOADS AS SHOWN IN THE ALTERNATE LOADING PLAN ON PAGE 19. THESE LOADING PROCEDURES CAN ALSO BE USED FOR A CARLOAD OF 45 UNIT LOADS BY SUBSTITUTING DUMMY LOADS (SEE PAGE 42) FOR UNIT LOADS IN THE UPPER LAYER OF EITHER STACK ADJACENT TO THE DOORWAY AREA. (LOADING PLAN AND CARLOAD DATA SHOWN ON PAGE 22).
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED. QUANTITIES CAN ALSO BE ADJUSTED BY SUBSTITUTING DUMMY UNIT LOADS (SEE PAGE 42) AS REQUIRED TO COMPLY WITH LOAD LIMIT OR WEIGHT DISTRIBUTION RULES OF THE A.A.R.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING THE LENGTH OF PIECE 18 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 18 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS. DOORWAY PROTECTION IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.
5. THE SWAY BRACE FRAME SHOWN IN DETAIL A IS FOR BOXCARS WITH 9'-2" OR 9'-4" INSIDE WIDTH. FOR BOXCARS WITH LESS THAN 9'-2" WIDTH, PIECE 6 OF DETAIL A SHOULD BE 2 x 4 MATERIAL AND FOR BOXCARS WIDER THAN 9'-4" PIECE 6 SHOULD BE 2 x 8 MATERIAL.

*2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

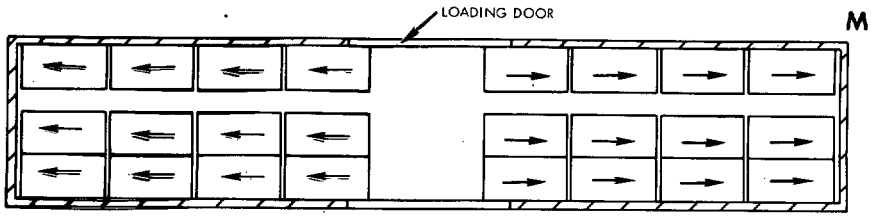


DETAIL A
SWAY BRACE FRAME
8 REQUIRED

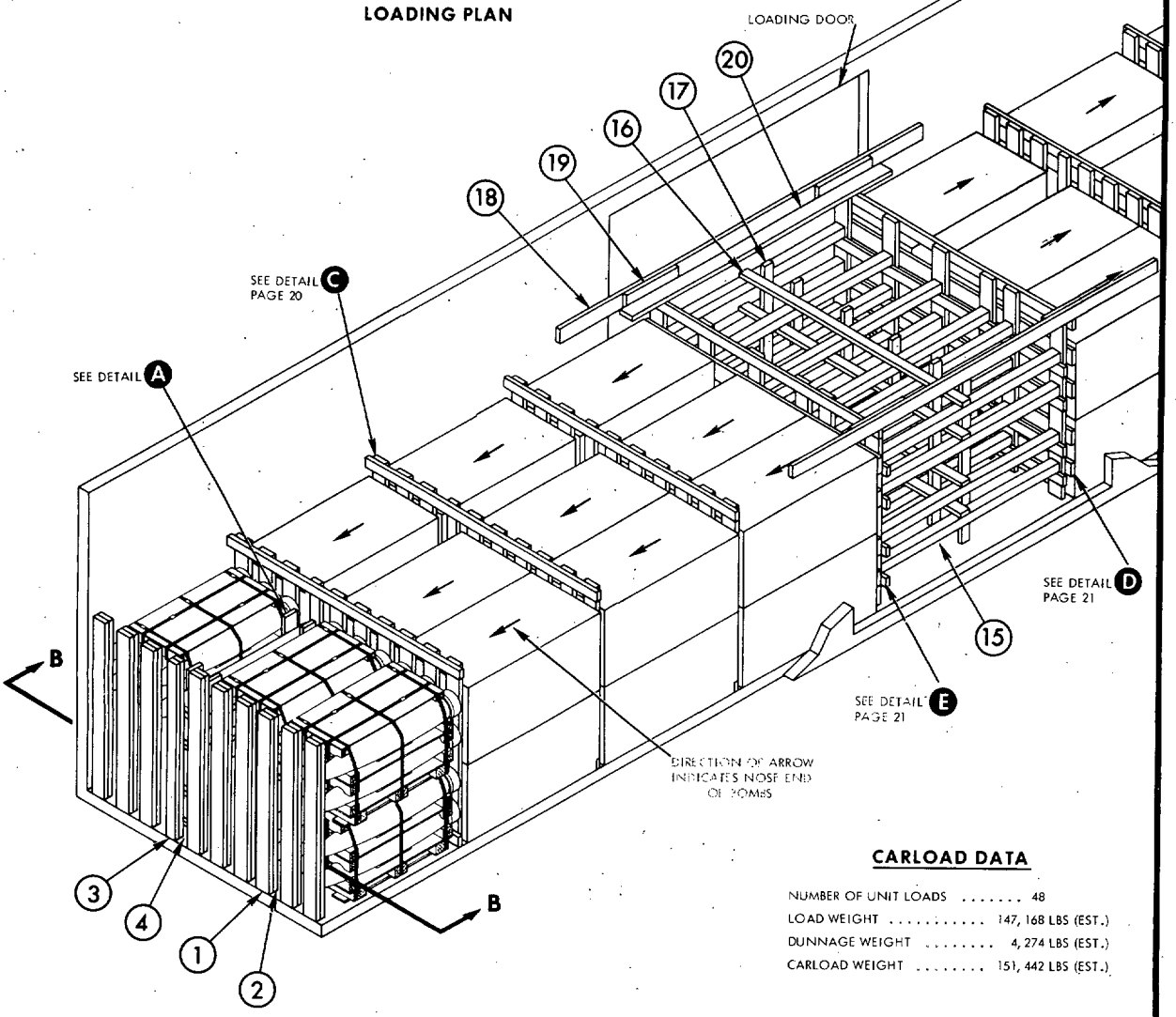
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE NAILS
20	CENTER GATE SWAY BRACE	2 x 6 x 10 FT	2	13	3 PER JOINT	16d
19	HOLD DOWN CLEAT	2 x 6 x 24	4	18	5	10d
18	GATE HOLD DOWN	2 x 6 x 14 FT	2	CAR WALL	5 EACH END	10d
17	VERTICAL TIE BAR	2 x 4 x 66	6	15	3 PER JOINT	16d
16	HORIZONTAL TIE BAR	2 x 4 x CAR WIDTH-1	4	15	3 PER JOINT	16d
15	STRUT	4 x 4 x WEDGE FIT*	24	11	2 PER JOINT	16d
14	SWAY BRACE STOP	2 x 6 x 30	2	11	3 PER JOINT	10d
13	CENTER GATE HORIZONTAL	2 x 4 x CAR WIDTH-1	2	11	3 PER JOINT	10d
12	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	11	3 PER JOINT	10d
11	CENTER GATE VERTICAL	2 x 6 x 72	12	SEE 10	-	-
10	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	11	3 PER JOINT	10d
9	SWAY BRACE STOP	2 x 4 x 72	6	SEE 8	-	-
8	SEPARATOR GATE HORIZONTAL	2 x 4 x CAR WIDTH-1	12	7,9	3 PER JOINT	10d
7	SEPARATOR GATE VERTICAL	2 x 6 x 72	54	SEE 8	-	-
6	CROSS BRACE	2 x 6 x 61 1/4	32	SEE 5	-	-
5	SWAY BRACE VERTICAL	2 x 6 x 60	32	6	3 PER JOINT	16d
4	END WALL MEMBER	2 x 4 x 72	2	3	1 PER FOOT	10d
3	END WALL MEMBER	2 x 4 x 72	2	CAR WALL	1 PER FOOT	10d
2	END WALL MEMBER	2 x 6 x 72	18	1	1 PER FOOT	10d
1	END WALL MEMBER	2 x 6 x 72	18	CAR WALL	1 PER FOOT	10d

LIST OF MATERIALS AND NAILING DATA

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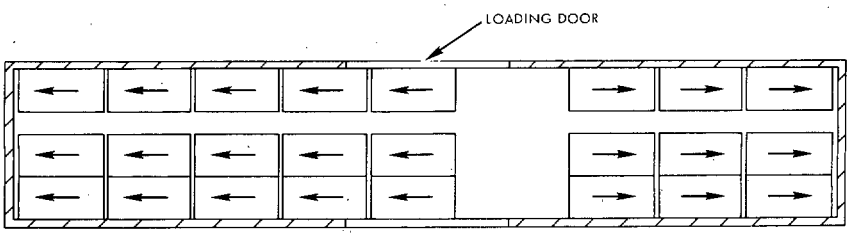


LOADING PLAN



CARLOAD DATA

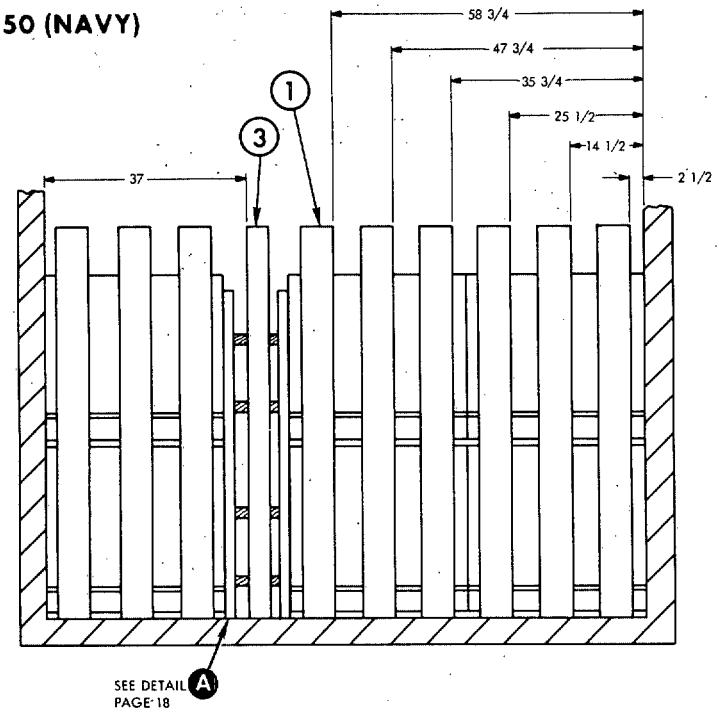
NUMBER OF UNIT LOADS	48
LOAD WEIGHT	147, 168 LBS (EST.)
DUNNAGE WEIGHT	4, 274 LBS (EST.)
CARLOAD WEIGHT	151, 442 LBS (EST.)



ALTERNATE LOADING PLAN

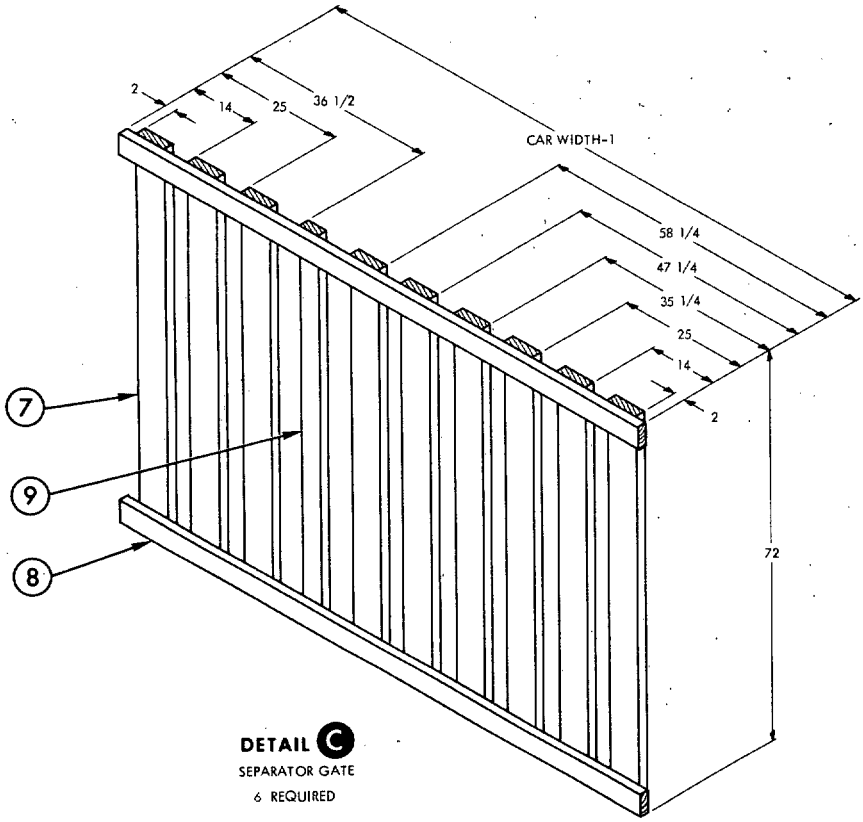
(SEE NOTE 1)

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SECTION B-B

SHOWING LOCATION OF END WALL MEMBERS
(REVERSE DIMENSIONS FOR OTHER END OF CAR)

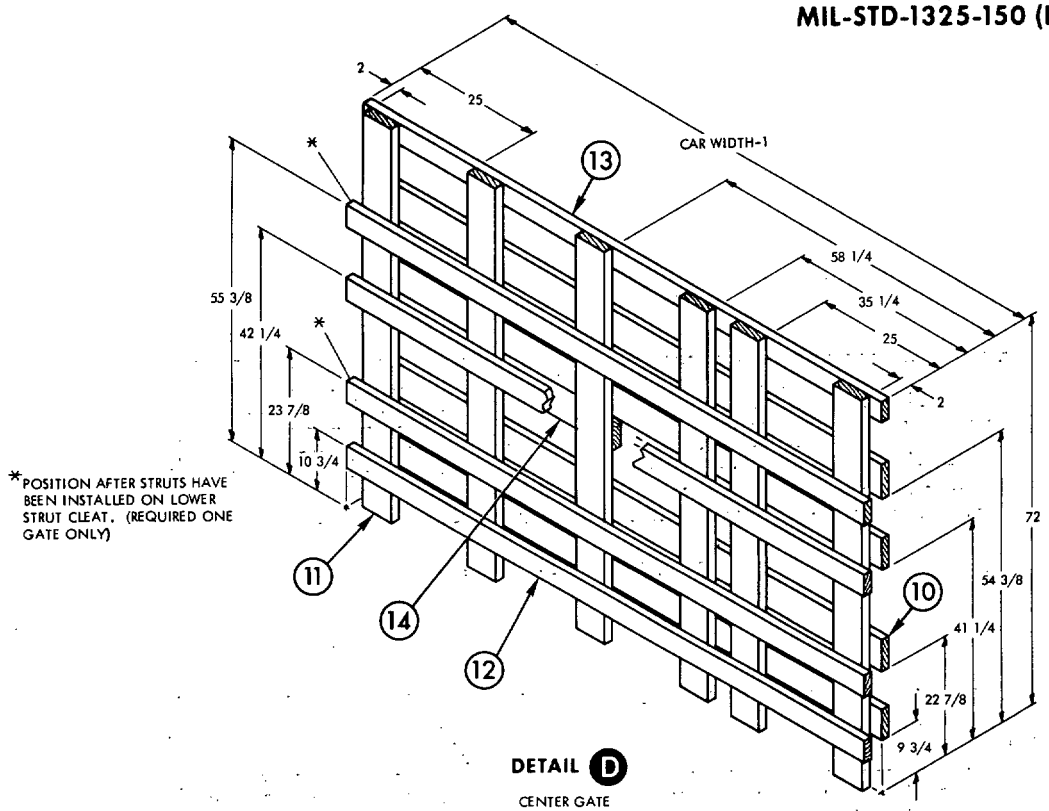


DETAIL C

SEPARATOR GATE

6 REQUIRED

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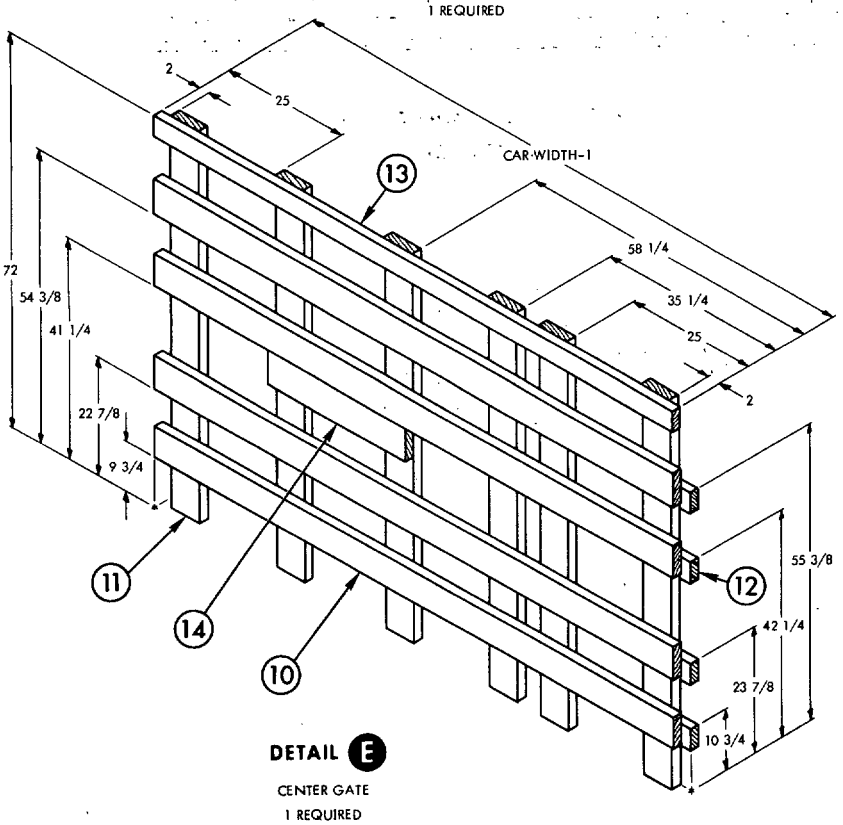


*POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT. (REQUIRED ONE GATE ONLY)

DETAIL D

CENTER GATE

1 REQUIRED

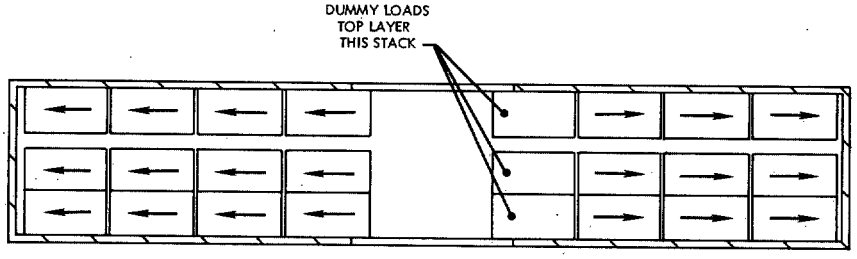


DETAIL E

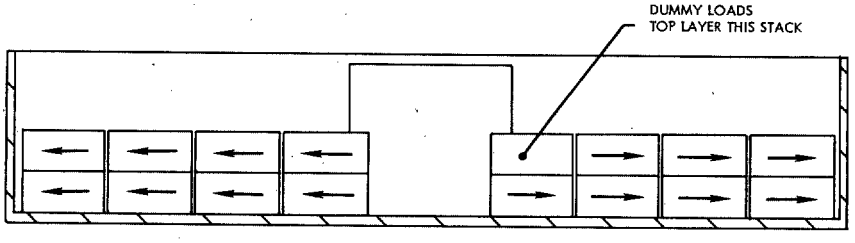
CENTER GATE

1 REQUIRED

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ALTERNATE LOADING PLAN



ALTERNATE LOADING PLAN
ELEVATION

CARLOAD DATA

NUMBER OF UNIT LOADS	45
LOAD WEIGHT	137, 970 LBS (EST.)
DUNNAGE WEIGHT	4, 874 LBS (EST.)
CARLOAD WEIGHT	142, 844 LBS (EST.)

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50 FT 6 IN BOXCAR, COMMERCIAL (ALTERNATE METHOD)

1. THE CARLOAD CONSISTS OF EITHER 33 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING OR 27 UNIT LOADS COMPRISING THE FIRST LAYER ONLY OF THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL FLOOR LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. QUANTITIES CAN ALSO BE ADJUSTED BY SUBSTITUTING DUMMY UNIT LOADS (SEE PAGE 42) AS REQUIRED TO COMPLY WITH LOAD LIMIT OR WEIGHT DISTRIBUTION RULES OF THE A.A.R.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING THE LENGTH OF PIECE 29 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 29 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS. DOORWAY PROTECTION, PIECES 31 AND 32 IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.
5. THE SWAY BRACE FRAME SHOWN IN DETAIL A IS FOR BOXCARS WITH 9'-2" OR 9'-4" INSIDE WIDTH. FOR BOXCARS WITH LESS THAN 9'-2" WIDTH, PIECE 6 OF DETAIL A SHOULD BE 2 x 4 MATERIAL AND FOR BOXCARS WIDER THAN 9'-4" PIECE 6 SHOULD BE 2 x 8 MATERIAL.

*2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

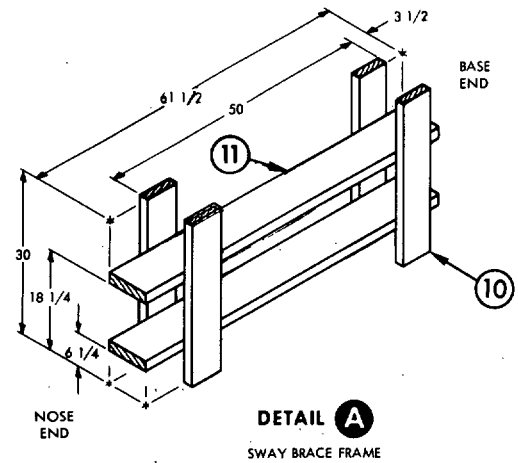
LIST OF MATERIALS AND NAILING DATA

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
32	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	6	31	2 PER JOINT	10d
31	VERTICAL DOORWAY MEMBER	2 x 3 x 36	4	DOOR POST	2 PER FOOT	20d
30	HOLD DOWN CLEAT	2 x 6 x 48	2	29	2 PER FOOT	10d
29	GATE HOLD DOWN	2 x 6 x 14 FT	2	CAR WALL	5 EACH END	10d
28	STRUT	4 x 4 x WEDGE FIT*	12	24	2 PER JOINT	16d
27	SWAY BRACE STOP	2 x 6 x 30	2	24	3 PER JOINT	10d
26	CENTER GATE HORIZONTAL	2 x 4 x CAR WIDTH-1	2	24	3 PER JOINT	10d
25	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	4	24	3 PER JOINT	10d
24	CENTER GATE VERTICAL	2 x 6 x 36	12	SEE 23	-	-
23	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	4	24	3 PER JOINT	10d
22	UPPER WALL CLEAT	2 x 4 x 18	4	CAR WALL	4	10d
21	VERTICAL BACK-UP CLEAT	2 x 6 x 36	4	CAR WALL	8	10d
20	INTERMEDIATE WALL CLEAT	2 x 4 x 9 1/2	4	CAR WALL	4	10d
19	HORIZONTAL BACK-UP CLEAT	2 x 6 x 30	8	16	14	16d
18	DIAGONAL BRACE	4 x 4 x 50	8	14,16	1 EACH END	60d
17	HORIZONTAL POCKET CLEAT	2 x 6 x 18	8	16	7	16d
16	HORIZONTAL WALL CLEAT	2 x 6 x 72	8	CAR WALL	16	10d

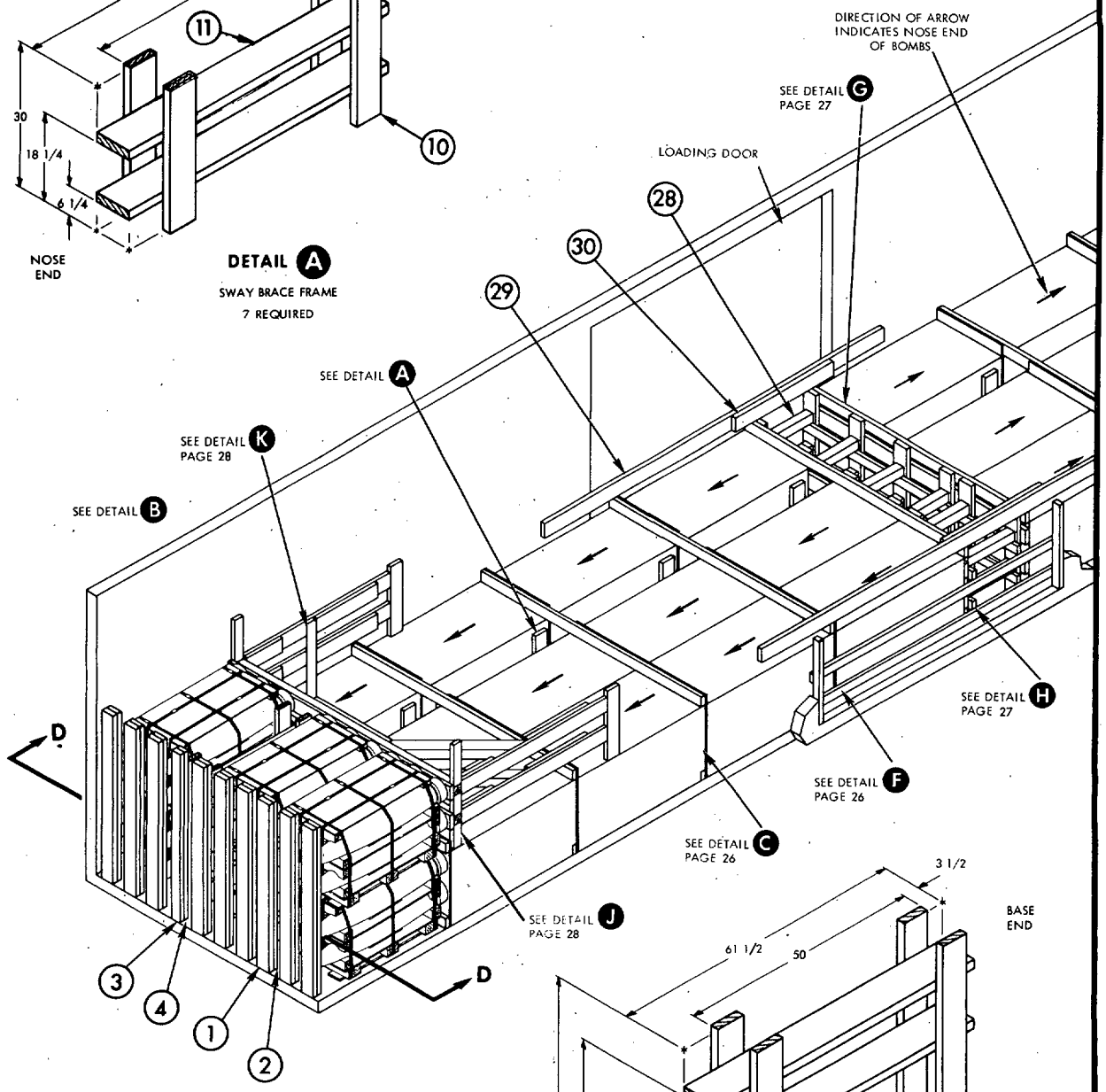
LIST OF MATERIALS AND NAILING DATA

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
15	CENTER CLEAT	2 x 4 x 36	4	14	7	16d
14	CROSS BRACE	4 x 4 x CAR WIDTH	4	SEE 13	-	-
13	CROSS BRACE STIFFENER	2 x 6 x CAR WIDTH	4	14	2 PER FOOT	16d
12	LOWER WALL CLEAT	2 x 4 x 10 5/8	4	CAR WALL	3	10d
11	CROSS BRACE	2 x 6 x 61 1/2	14	SEE 10	-	-
10	SWAY BRACE VERTICAL	2 x 6 x 30	28	11	3 PER JOINT	16d
9	TIE PIECE	2 x 4 x CAR WIDTH-1	14	SEE 7	-	-
8	PLYWOOD SPACER	1/2 x 36 x 32	7	9	2 PER FOOT	6d
7	PLYWOOD SPACER	1/2 x 36 x 64	7	9	2 PER FOOT	6d
6	CROSS BRACE	2 x 6 x 61 1/2	8	SEE 5	-	-
5	SWAY BRACE VERTICAL	2 x 6 x 60	8	6	3 PER JOINT	16d
4	END WALL MEMBER	2 x 4 x 72	2	3	1 PER FOOT	10d
3	END WALL MEMBER	2 x 4 x 72	2	CAR WALL	1 PER FOOT	10d
2	END WALL MEMBER	2 x 6 x 72	18	1	1 PER FOOT	10d
1	END WALL MEMBER	2 x 6 x 72	18	CAR WALL	1 PER FOOT	10d

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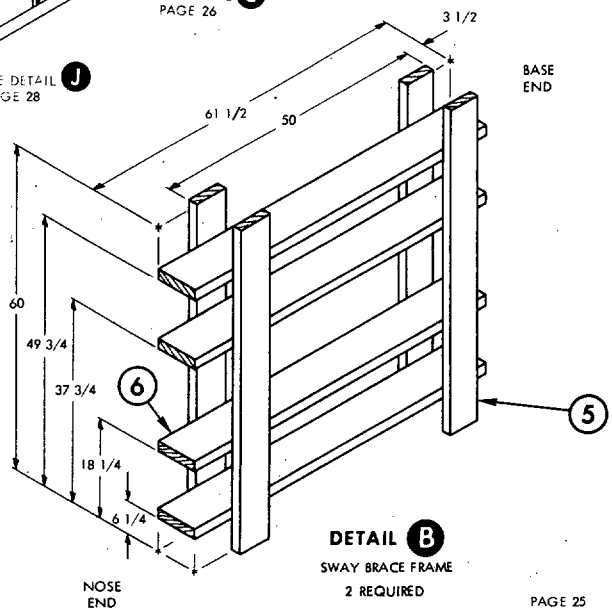


DETAIL A
SWAY BRACE FRAME
7 REQUIRED



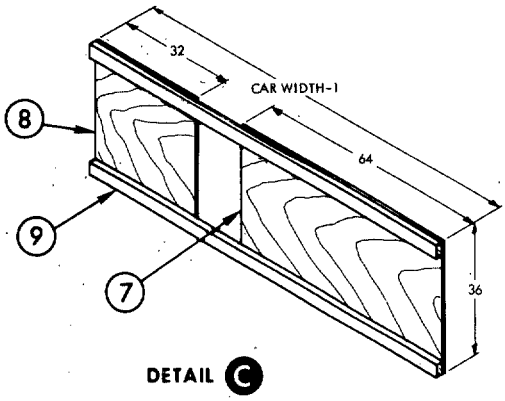
CARLOAD DATA

NUMBER OF UNIT LOADS	33
LOAD WEIGHT	101, 178 LBS (EST.)
DUNNAGE WEIGHT	2, 973 LBS (EST.)
CARLOAD WEIGHT	104, 151 LBS (EST.)



DETAIL B
SWAY BRACE FRAME
2 REQUIRED

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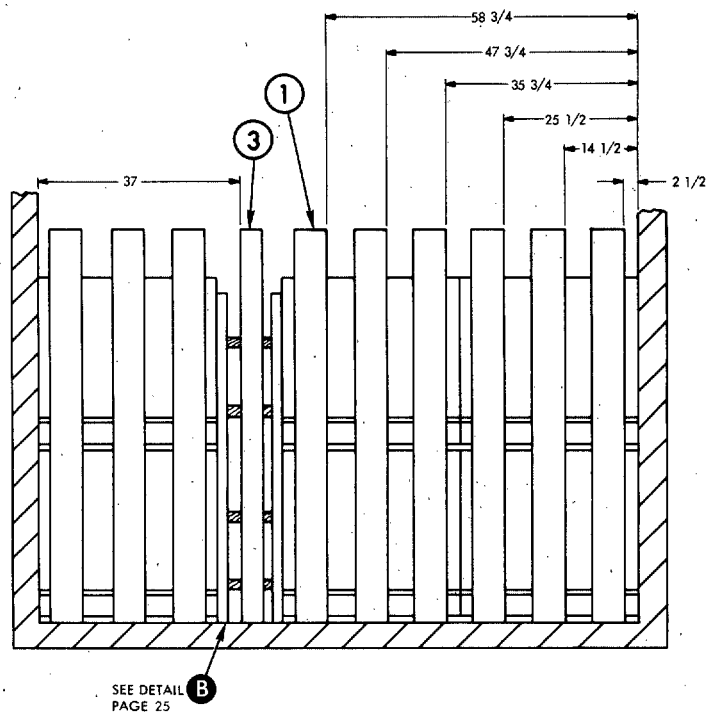


DETAIL C

SEPARATOR GATE

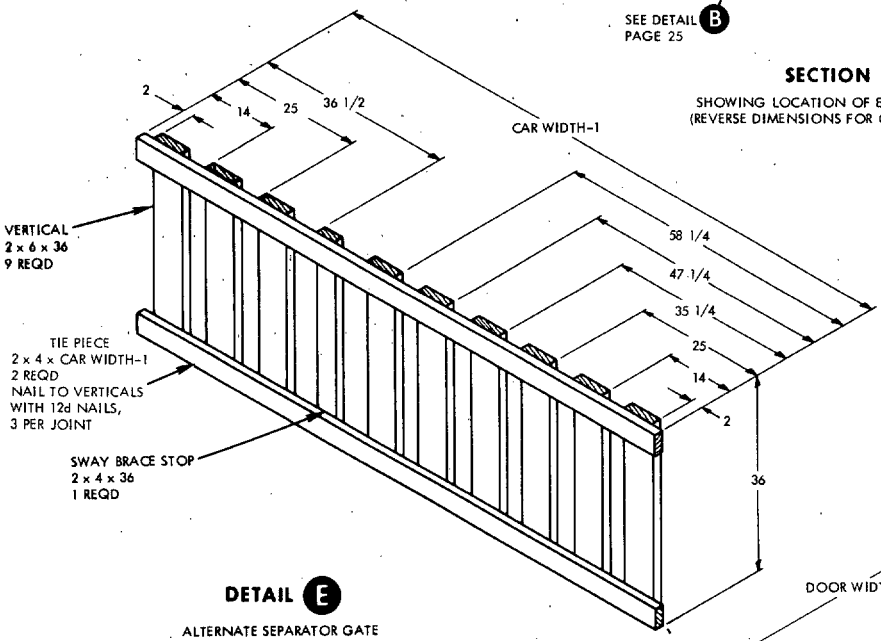
7 REQUIRED

(SEE ALTERNATE SEPARATOR GATE, DETAIL E)



SECTION D-D

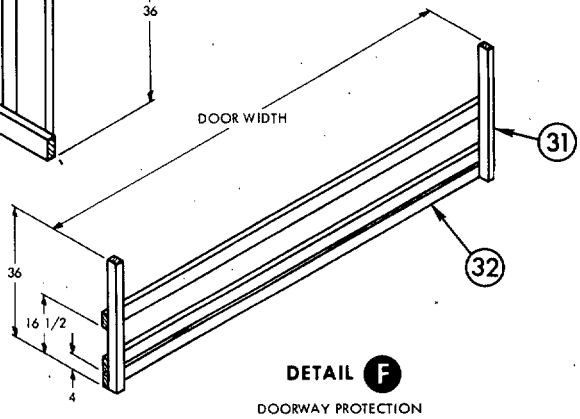
SHOWING LOCATION OF END WALL MEMBERS
(REVERSE DIMENSIONS FOR OTHER END OF CAR)



DETAIL E

ALTERNATE SEPARATOR GATE

7 REQUIRED



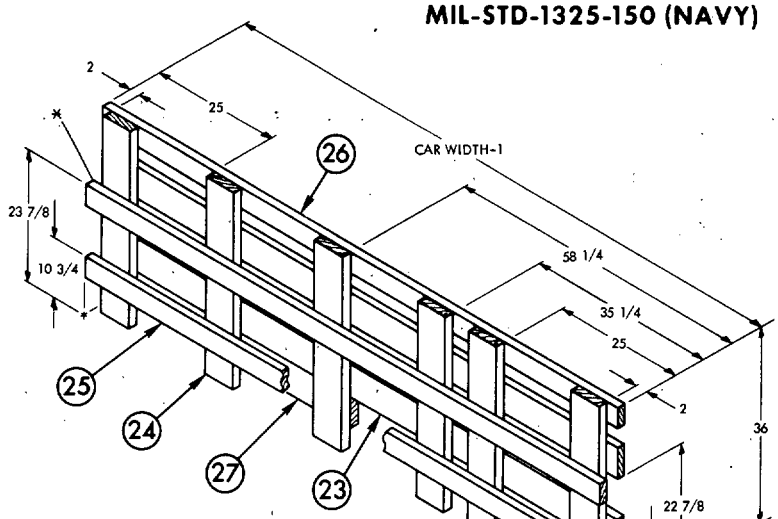
DETAIL F

DOORWAY PROTECTION

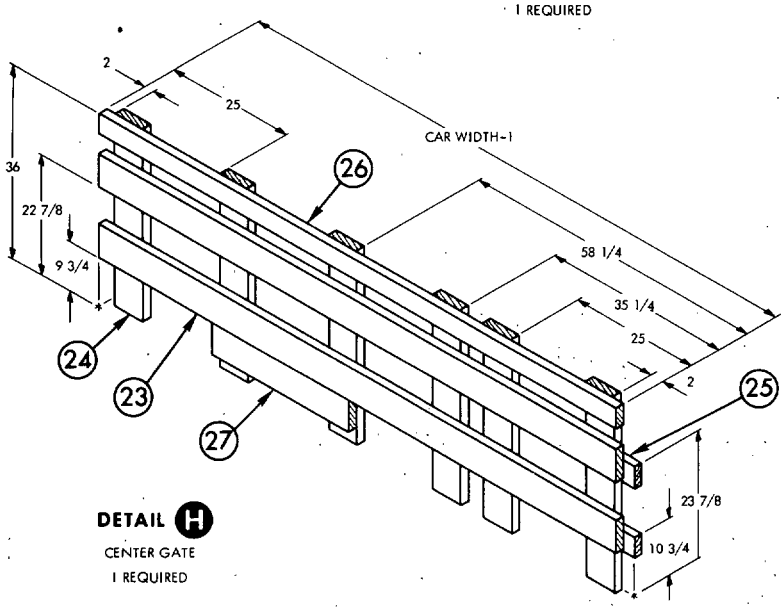
2 REQUIRED

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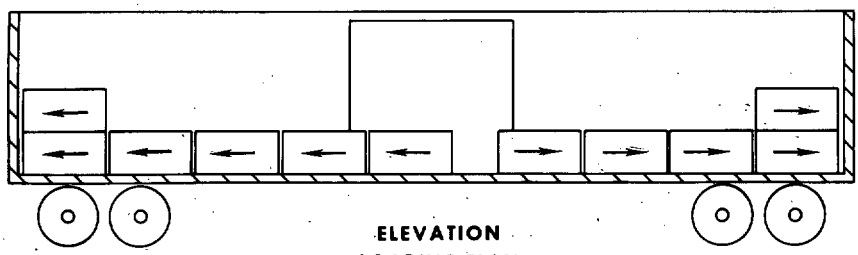
*POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT. (REQUIRED ONE GATE ONLY)



DETAIL G
CENTER GATE
1 REQUIRED

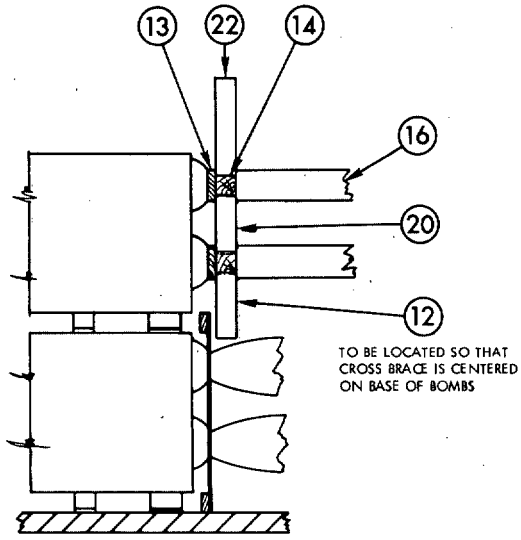


DETAIL H
CENTER GATE
1 REQUIRED

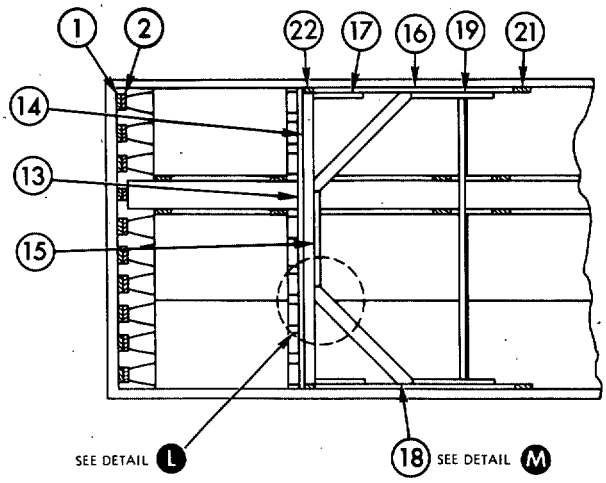


ELEVATION
LOADING PLAN

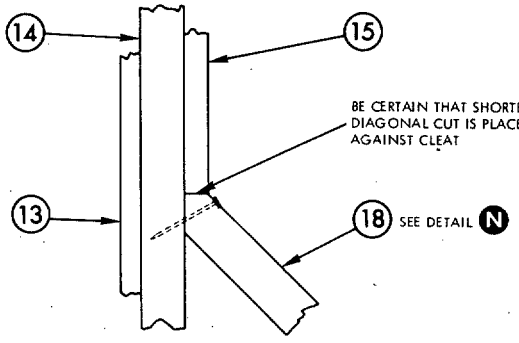
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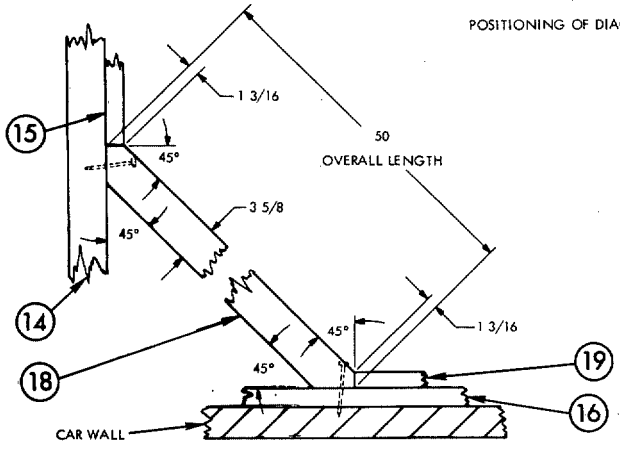
DETAIL J



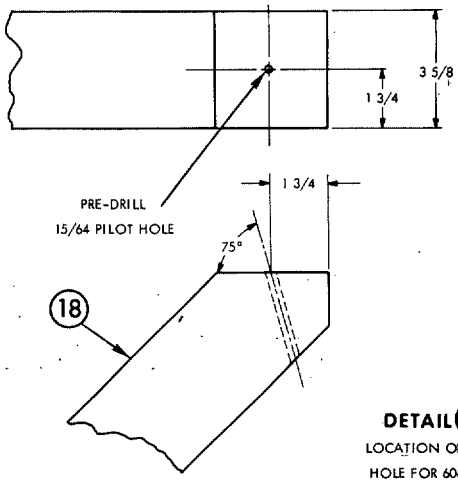
DETAIL K
PARTIAL LAYER BRACING



DETAIL L
POSITIONING OF DIAGONAL BRACE



DETAIL M
POSITIONING OF DIAGONAL BRACE



DETAIL N
LOCATION OF PILOT HOLE FOR 60d NAIL

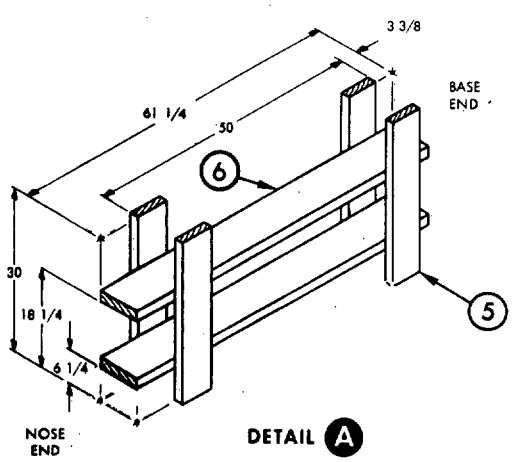
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50 FT 6 IN. BOXCAR, COMMERCIAL (ALTERNATE METHOD)

1. THE CARLOAD CONSISTS OF EITHER 45 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING OR 40 UNIT LOADS AS SHOWN IN ALTERNATE LOADING PLAN PAGE 33. FOR THE ALTERNATE LOADING PLAN CENTER GATE AREA SHALL BE DUNNAGED SIMILAR TO PROCEDURE SHOWN ON CARLOADING METHOD PAGE 19 EXCEPT FOR OMITTING THE FOUR UPPER LAYER MIDDLE ROW STRUTS.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED. QUANTITIES CAN ALSO BE ADJUSTED BY SUBSTITUTING DUMMY UNIT LOADS (SEE PAGE 42) AS REQUIRED TO COMPLY WITH LOAD LIMIT OR WEIGHT DISTRIBUTION RULES OF THE A.A.R.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING THE LENGTH OF PIECE 19 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 19 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS. DOORWAY PROTECTION, PIECES 21 AND 22 IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.
5. THE SWAY BRACE FRAME SHOWN IN DETAIL A IS FOR BOXCARS WITH 9'-2" OR 9'-4" INSIDE WIDTH. FOR BOXCARS WITH LESS THAN 9'-2" WIDTH, PIECE 6 OF DETAIL A SHOULD BE 2 x 4 MATERIAL AND FOR BOXCARS WIDER THAN 9'-4" PIECE 6 SHOULD BE 2 x 8 MATERIAL.



DETAIL A

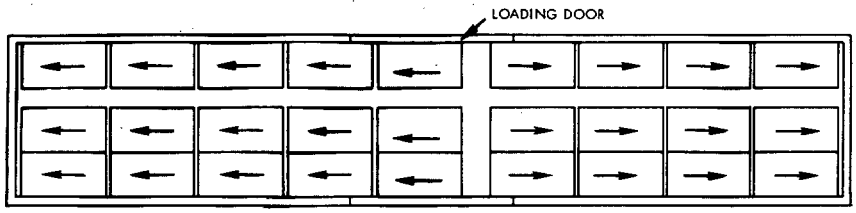
SWAY BRACE FRAME
9 REQUIRED

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE
22	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	10	21	2 PER JOINT	10d
21	VERTICAL DOORWAY MEMBER	2 x 3 x 72	4	DOOR POST	2 PER FOOT	20d
20	HOLD DOWN CLEAT	2 x 6 x 60	2	19	2 PER FOOT	10d
19	GATE HOLD DOWN	2 x 6 x 14 FT	2	CAR WALL	5 EACH END	10d
18	STRUT	4 x 4 x WEDGE FIT*	20	14	2 PER JOINT	16d
17	SWAY BRACE STOP	2 x 6 x 30	2	14	3 PER JOINT	10d
16	CENTER GATE HORIZONTAL	2 x 4 x CAR WIDTH-1	2	14	3 PER JOINT	10d
15	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	14	3 PER JOINT	10d
14	CENTER GATE VERTICAL	2 x 6 x 72	12	SEE 13	-	-
13	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	14	3 PER JOINT	10d
12	SWAY BRACE STOP	2 x 4 x 72	7	SEE 11	-	-
11	SEPARATOR GATE HORIZONTAL	2 x 4 x CAR WIDTH-1	14	10,12	3 PER JOINT	10d
10	SEPARATOR GATE VERTICAL	2 x 6 x 72	63	SEE 11	-	-
9	SWAY BRACE UPRIGHT	2 x 4 x 27 1/4	36	8	3 PER JOINT	10d
8	CROSS BRACE	2 x 4 x CUT TO FIT	72	SEE 7	-	-
7	SWAY BRACE LONGITUDINAL	2 x 4 x 61	36	8,9	2 PER JOINT	16d
6	CROSS BRACE	2 x 6 x 61 1/4	18	SEE 5	-	-
5	SWAY BRACE VERTICAL	2 x 6 x 30	36	6	3 PER JOINT	16d
4	END WALL MEMBER	2 x 4 x 72	2	3	1 PER FOOT	10d
3	END WALL MEMBER	2 x 4 x 72	2	CAR WALL	1 PER FOOT	10d
2	END WALL MEMBER	2 x 6 x 72	18	1	1 PER FOOT	10d
1	END WALL MEMBER	2 x 6 x 72	18	CAR WALL	1 PER FOOT	10d

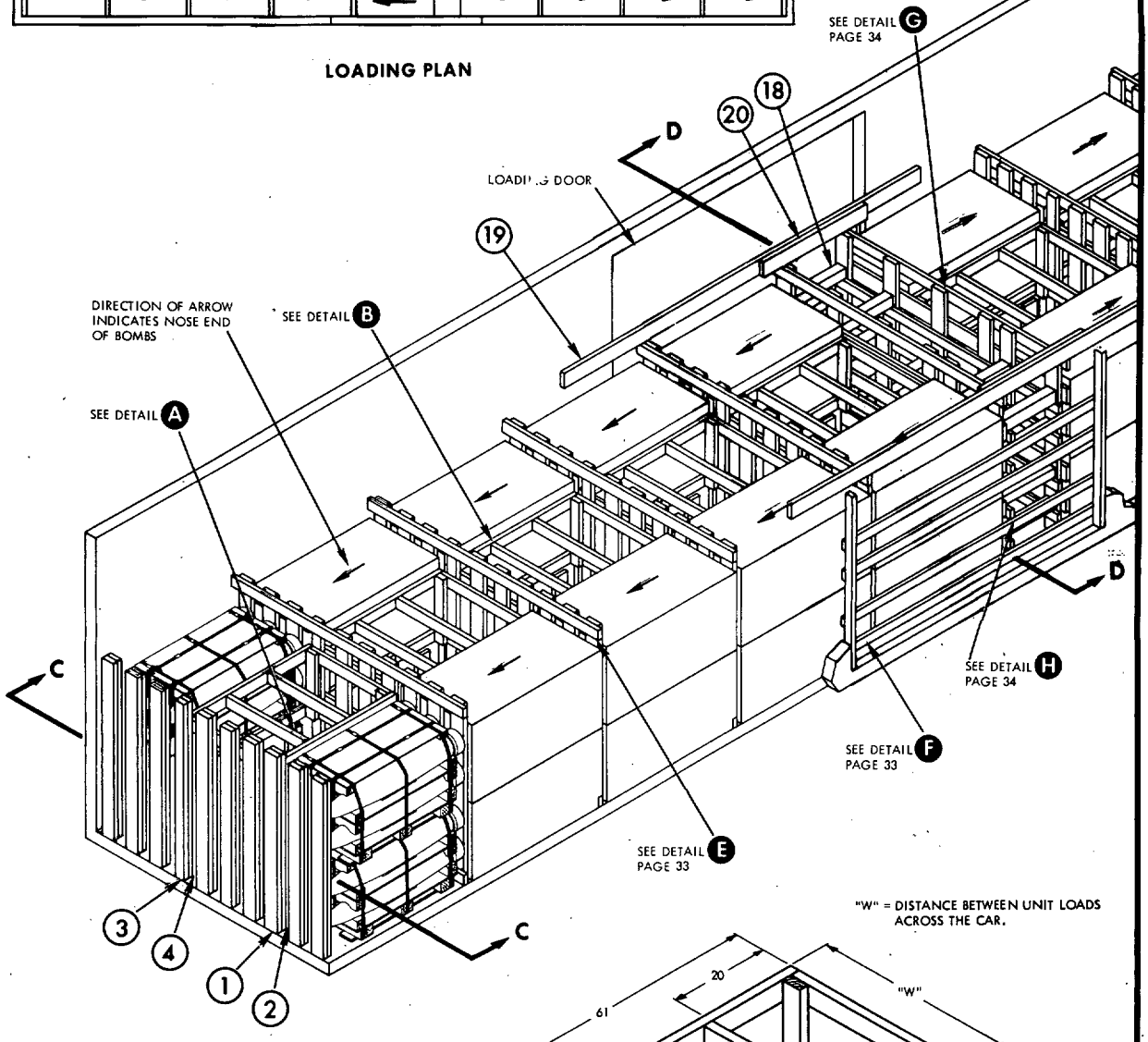
LIST OF MATERIALS AND NAILING DATA

*2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

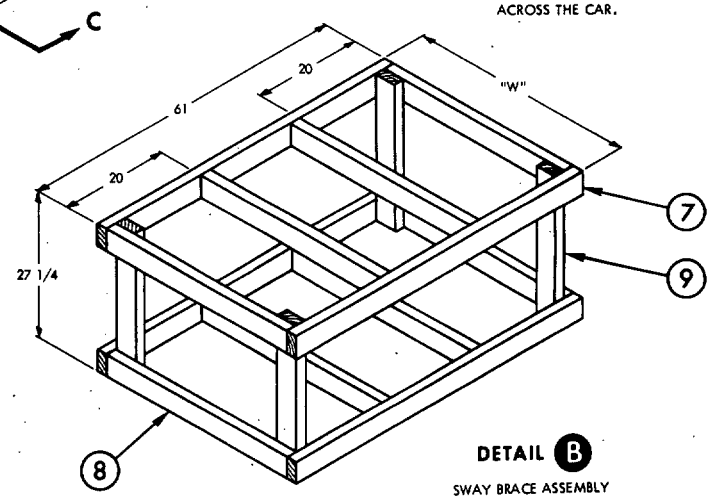
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LOADING PLAN



"W" = DISTANCE BETWEEN UNIT LOADS ACROSS THE CAR.



DETAIL B

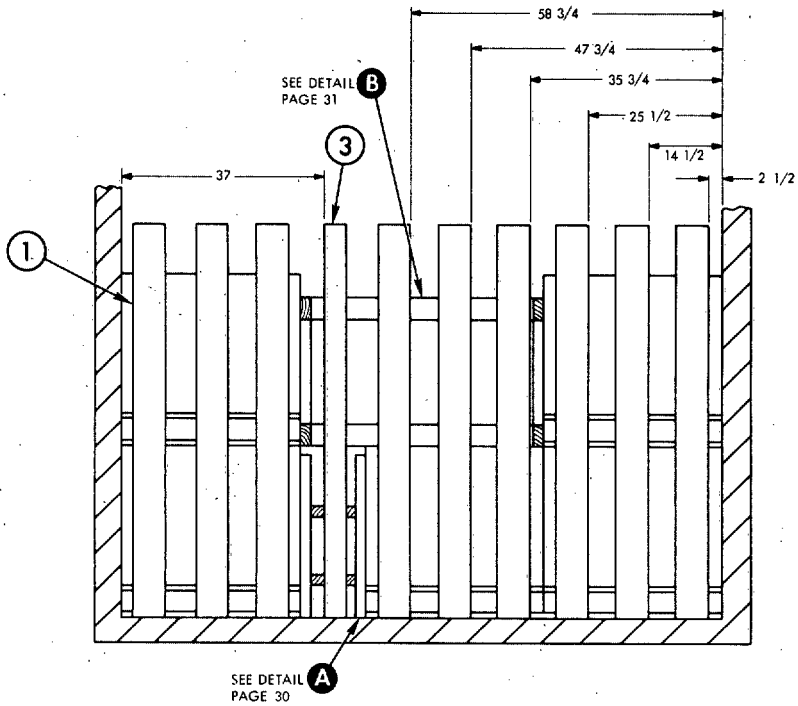
SWAY BRACE ASSEMBLY

9 REQUIRED

CARLOAD DATA

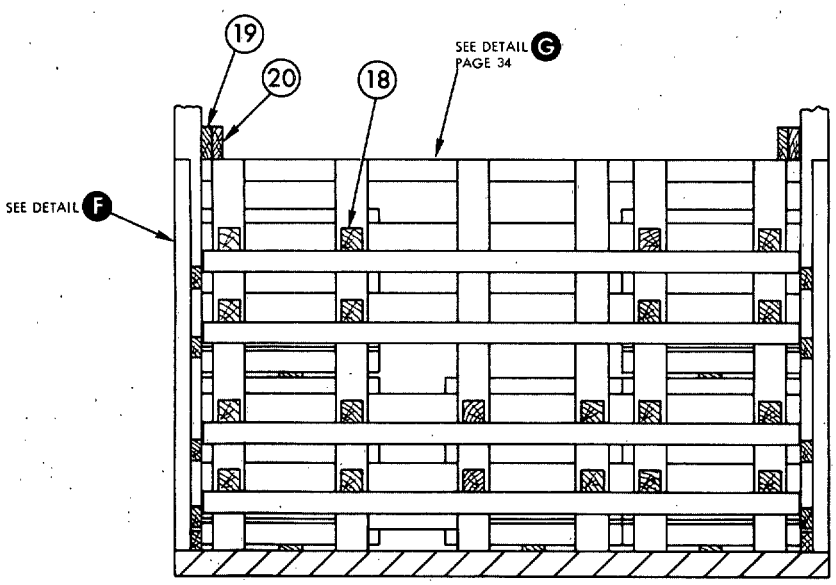
NUMBER OF UNIT LOADS	45
LOAD WEIGHT	137, 970 LBS (EST.)
DUNNAGE WEIGHT	4, 244 LBS (EST.)
CARLOAD WEIGHT	142, 214 LBS (EST.)

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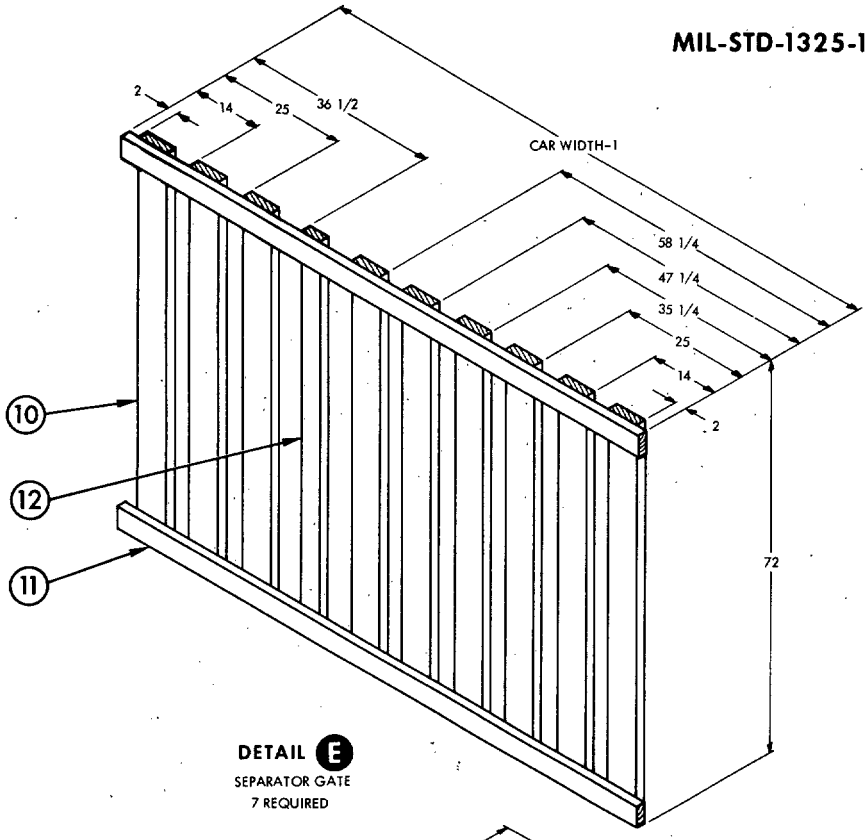
SECTION C-C

SHOWING LOCATION OF END WALL MEMBERS
(REVERSE DIMENSIONS FOR OTHER END OF CAR)

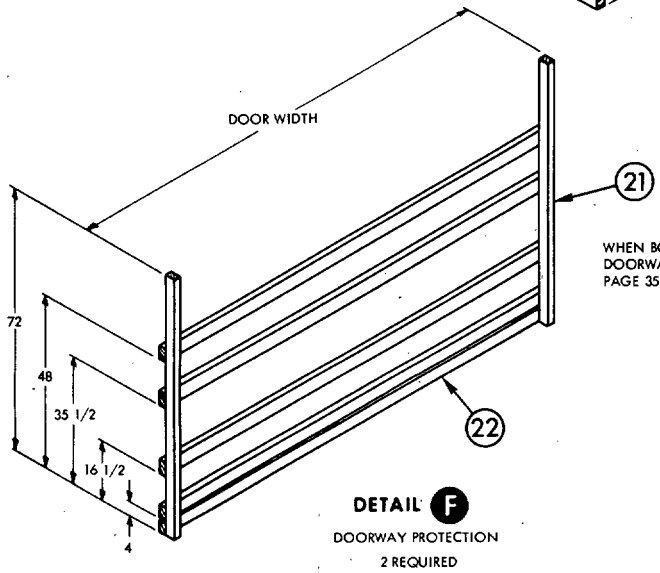


SECTION D-D

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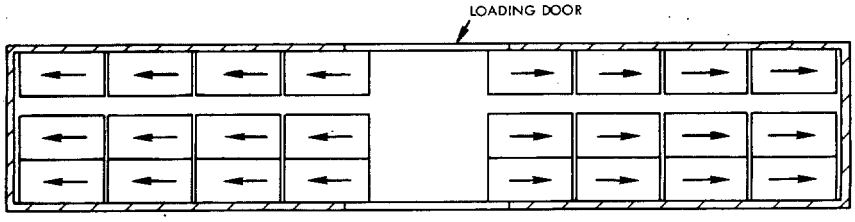


DETAIL E
SEPARATOR GATE
7 REQUIRED



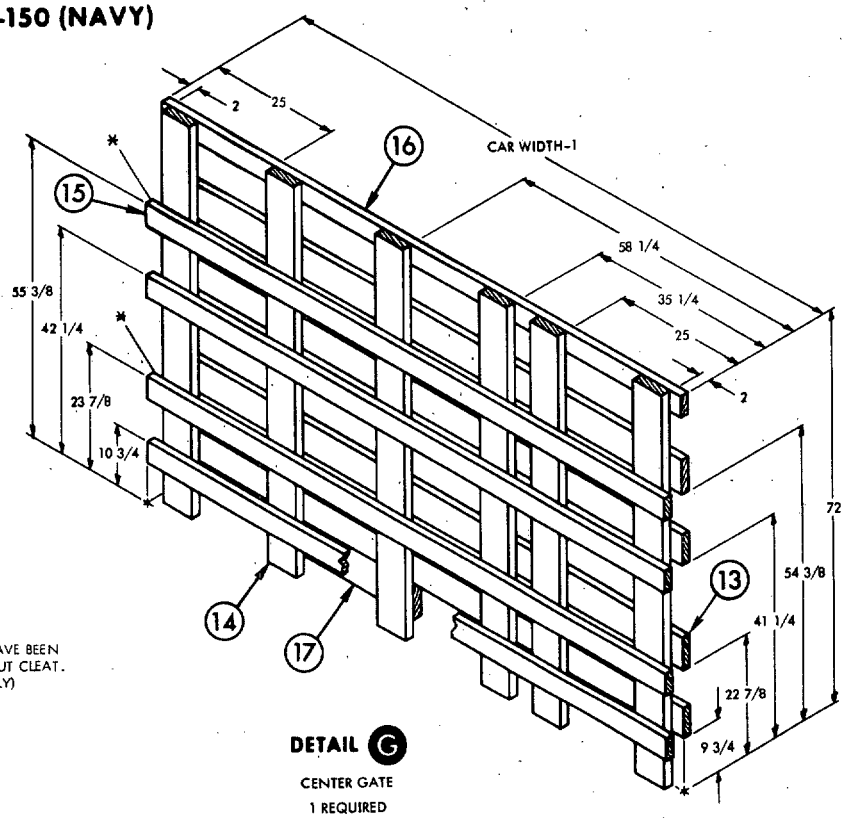
WHEN BOXCAR HAS STEEL DOOR POSTS, DOORWAY PROTECTION AS PER WR-52, PAGE 35, FIG. 21 IS REQUIRED.

DETAIL F
DOORWAY PROTECTION
2 REQUIRED



ALTERNATE LOADING PLAN

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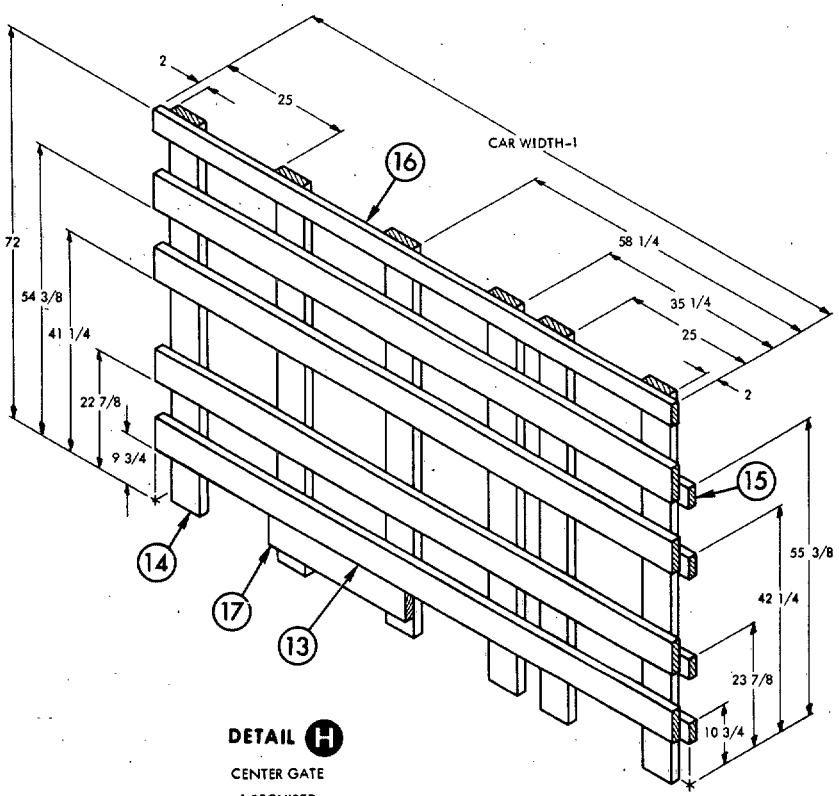


*POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT. (REQUIRED ONE GATE ONLY)

DETAIL G

CENTER GATE

1 REQUIRED



DETAIL H

CENTER GATE

1 REQUIRED

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50 FT 6 IN. BOXCAR, COMMERCIAL (USING LOAD DIVIDERS)

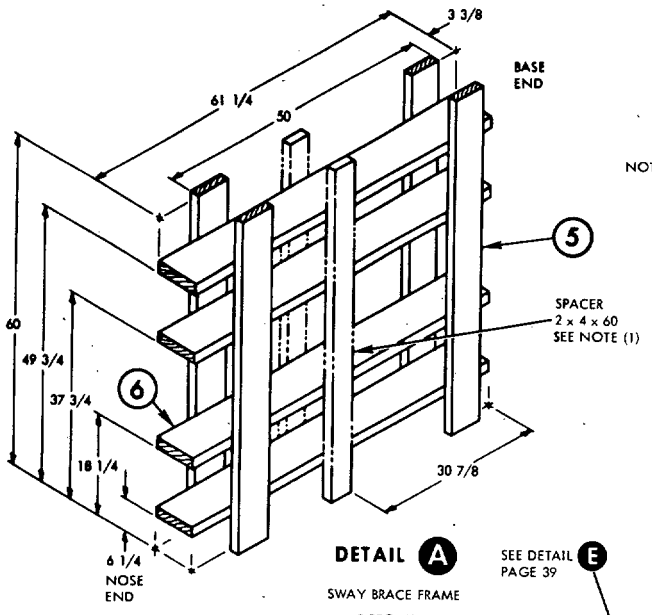
1. THESE PROCEDURES DEPICT THE METHOD OF LOADING THE AIR FORCE UNIT LOAD OF MK 82 BOMBS IN CUSHIONED BOXCARS WHICH ARE EQUIPPED WITH LOAD DIVIDERS AND WITH OR WITHOUT ADJUSTABLE SIDE FILLERS. ONLY CUSHIONED CARS THAT HAVE SLIDING CENTER SILL TYPE CUSHIONING DEVICES OR END-OF-CAR TYPE DEVICES WHICH HAVE AT LEAST 15 INCHES OF TRAVEL ARE ACCEPTABLE. (ONLY THOSE CUSHIONED BOXCARS EQUIPPED WITH LOAD DIVIDERS MANUFACTURED BY EVANS, EQUIPCO AND PRECO HAVE BEEN TESTED AND APPROVED BY THE AAR, BUX.). THE LOAD DIVIDERS WILL REPLACE THE CENTER GATE ASSEMBLY. A STRUT ASSEMBLY, DETAIL G, WILL BE REQUIRED BETWEEN THE LOAD DIVIDERS AS SHOWN IN THE LOAD VIEWS, IF EITHER LOAD DIVIDER IS REQUIRED TO RETAIN A LADING WEIGHT OF 50,000 POUNDS OR MORE.
2. BOXCARS EQUIPPED WITH ADJUSTABLE SIDE FILLERS THAT HAVE 3/8 INCH OR THICKER PANELS MAY BE USED. HOWEVER, THESE SIDE FILLERS MUST NOT BE USED FOR LATERAL BLOCKING. THEY MUST BE RETRACTED AND LOCKED AGAINST THE CAR SIDE WALL, AND A FILL PIECE INSTALLED IN THE VOID, IF ONE EXISTS, BETWEEN THE CAR SIDE WALL AND THE SIDE FILLER PANEL. (SEE DETAIL H, PAGE 40).
3. PRIOR TO LOADING A CAR EQUIPPED WITH LOAD DIVIDERS, A VERY CAREFUL INSPECTION MUST BE MADE TO ENSURE THAT THE CAR AND THE CAR EQUIPMENT IS IN GOOD CONDITION. THE CONDITION OF THE LOAD DIVIDERS SHOULD BE CHECKED THROUGHLY. BREAKS IN WELDING, BENT OR OTHERWISE DAMAGED LOCKING PINS, AND BENT OR DEFORMED BULKHEADS ARE REASONS FOR REJECTING CARS. AFTER THE LOAD DIVIDERS ARE POSITIONED AGAINST THE LADING, AND THE LOCKING PINS ARE ENGAGED IN THE HOLES OF THE RAILS, THE LOCKING PINS MUST BE INSPECTED TO ENSURE THAT THE PINS ARE FULLY ENGAGED IN THE LOCKING HOLES. IF THE LOWER PINS ARE NOT FULLY SEATED IN THE HOLES, THE LINKAGE MECHANISM SHOULD BE ADJUSTED AS REQUIRED SO THAT THE PINS WILL BE FULLY SEATED INTO THE LOCKING HOLES OF THE LOWER RAILS. IF PRESENT, DEBRIS MUST BE REMOVED FROM BENEATH THE LOCKING HOLES SELECTED FOR SECURING A DIVIDER.
4. THE CARLOAD CONSISTS OF 46 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING. THERE ARE 23 UNIT LOADS AND ONE DUMMY UNIT LOAD EACH END OF THE CAR. THE DUMMY UNIT LOAD IS REQUIRED TO FILL OUT THE VOID SPACE AND MUST BE CONSTRUCTED AS SHOWN ON PAGE 42. (IF LOAD LIMIT OF CAR PERMITS, REPLACE DUMMY LOADS WITH BOMB LOADS).
5. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS, ENTIRE STACKS SHALL BE ELIMINATED WHERE POSSIBLE, OTHERWISE ADDITIONAL DUMMY UNIT LOADS SHALL BE USED TO FILL OUT THE VOIDS IN A STACK.
6. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING DOORWAY PROTECTION IS INSTALLED WHEN REQUIRED. (SEE DETAIL F, PAGE 33).
7. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4 INCH STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS. DOORWAY PROTECTION IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.
8. WHEN BOXCARS OTHER THAN 9 FT 2 INCHES WIDE ARE USED, DUNNAGE NAILED TO THE SIDE WALL OF THE CAR MUST BE INSTALLED AS NECESSARY FOR A TIGHT LOAD ACROSS THE CAR.

* "W" = DISTANCE BETWEEN LOCKING PINS AT EACH SIDE OF THE LOAD DIVIDER.

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
17	TIE BAR	2 x 4 x CAR WIDTH - 1	1	16	3 PER JOINT	10d
16	STRUT	4 x 4 CUT TO FIT	4	15	2 EACH END	16d
15	BUFFER PIECE	2 x 4 x CAR WIDTH - 1	2	SEE 14	-	-
14	STRUT LEDGER	2 x 4 x CAR WIDTH - 1	2	15	1 PER FOOT	10d
13	HOLD DOWN	1 x 6 x "W"*	2	14	1 PER FOOT	6d
12	SWAY BRACE STOP	1 x 4 x 69 1/2	2	11	3 PER JOINT	6d
11	SEPARATOR GATE HORIZONTAL	2 x 6 x CAR WIDTH - 1	8	SEE 10	-	-
10	SEPARATOR GATE VERTICAL	1 x 6 x 69 1/2	18	11	3 PER JOINT	6d
9	SWAY BRACE STOP	2 x 4 x 72	6	SEE 8	-	-
8	SEPARATOR GATE HORIZONTAL	2 x 4 x CAR WIDTH - 1	12	7, 9	3 PER JOINT	10d
7	SEPARATOR GATE VERTICAL	2 x 6 x 72	54	SEE 8	-	-
6	CROSS BRACE	2 x 6 x 61 1/4	32	SEE 5	-	-
5	SWAY BRACE VERTICAL	2 x 6 x 60	32	6	3 PER JOINT	16d
4	END WALL MEMBER	2 x 4 x 72	2	3	1 PER FOOT	10d
3	END WALL MEMBER	2 x 4 x 72	2	CAR WALL	1 PER FOOT	10d
2	END WALL MEMBER	2 x 6 x 72	18	1	1 PER FOOT	10d
1	END WALL MEMBER	2 x 6 x 72	18	CAR WALL	1 PER FOOT	10d

LIST OF MATERIALS & NAILING DATA

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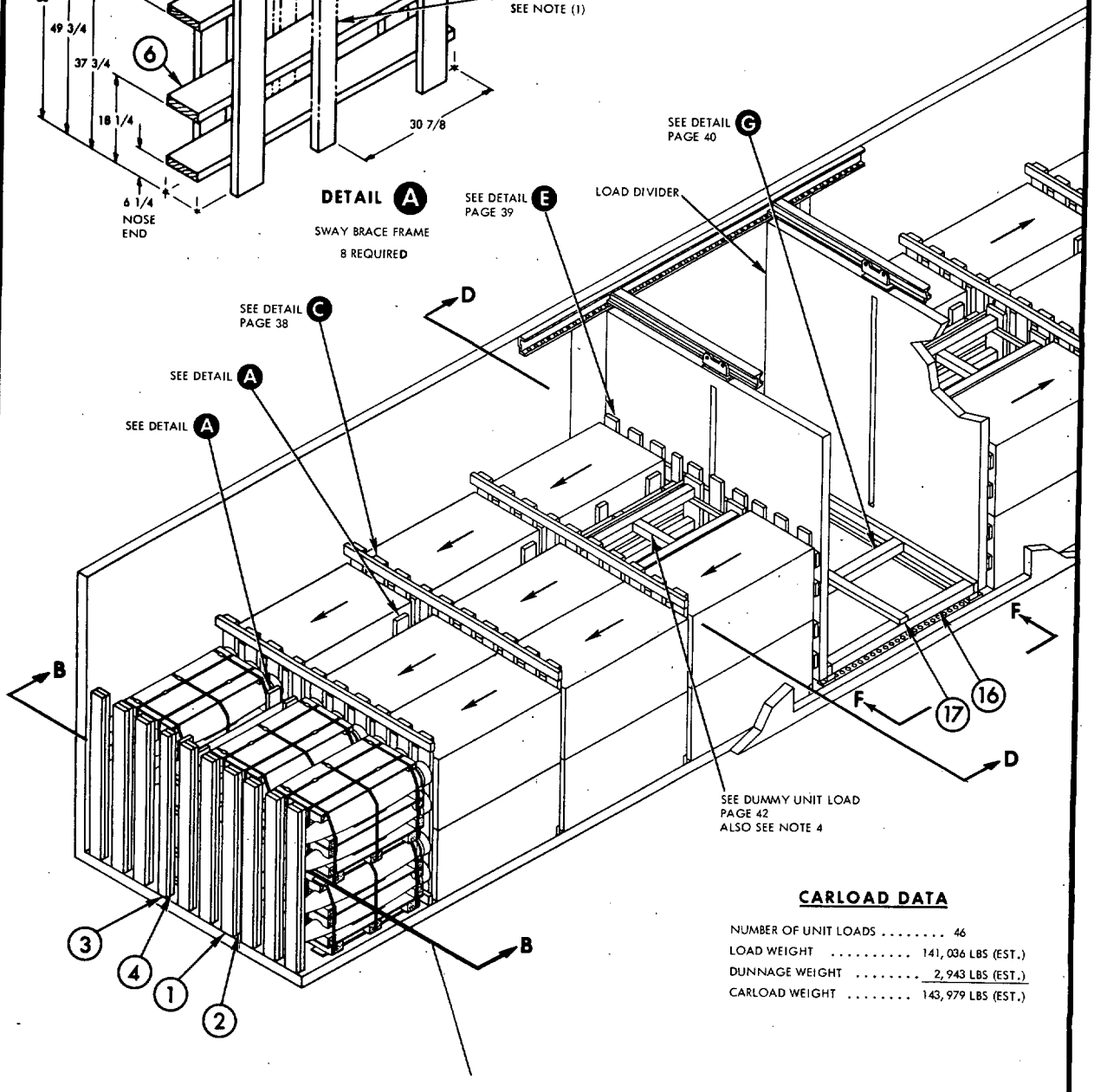
DETAIL A

SWAY BRACE FRAME

8 REQUIRED

NOTE:

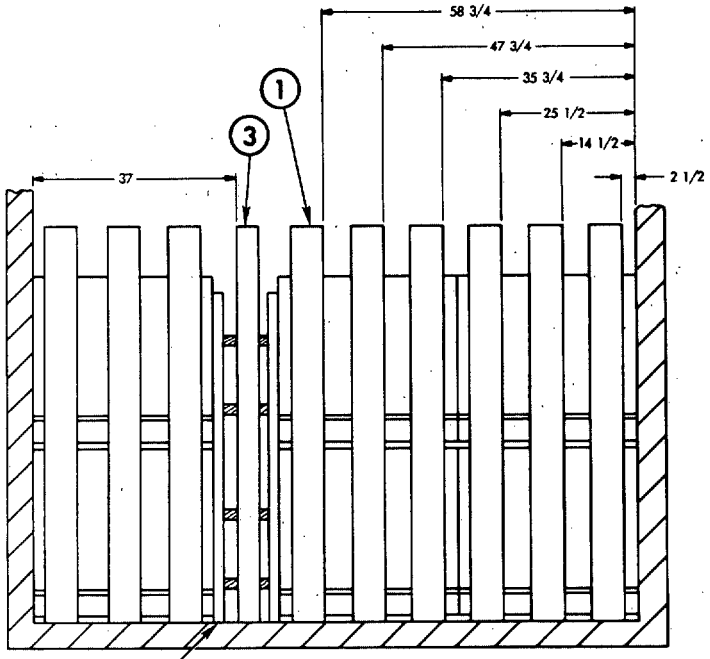
(1) WHEN POSITIONING DUMMY LOAD NEXT TO SWAY BRACE FRAME, 2 x 4 SPACERS MUST BE INSTALLED ON EACH SIDE OF FRAME. THIS IS NOT REQUIRED WHEN 3 DUMMY LOADS ARE LOADED ACROSS THE CAR.



CARLOAD DATA

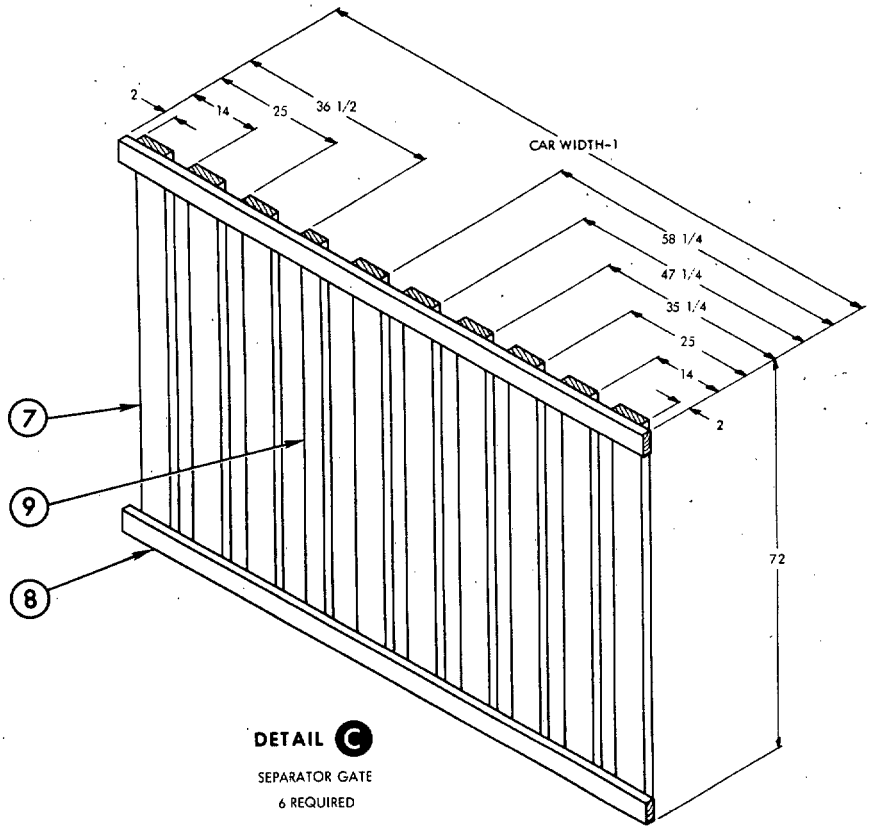
NUMBER OF UNIT LOADS	46
LOAD WEIGHT	141,036 LBS (EST.)
DUNNAGE WEIGHT	2,943 LBS (EST.)
CARLOAD WEIGHT	143,979 LBS (EST.)

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SECTION B-B

SHOWING LOCATION OF END WALL MEMBERS

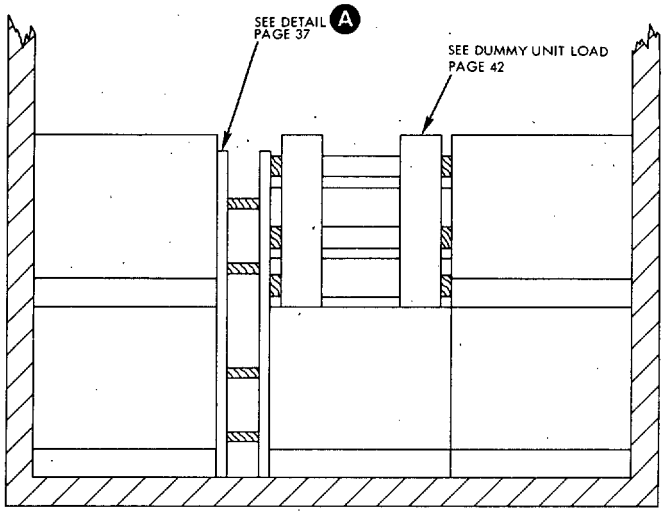


DETAIL C

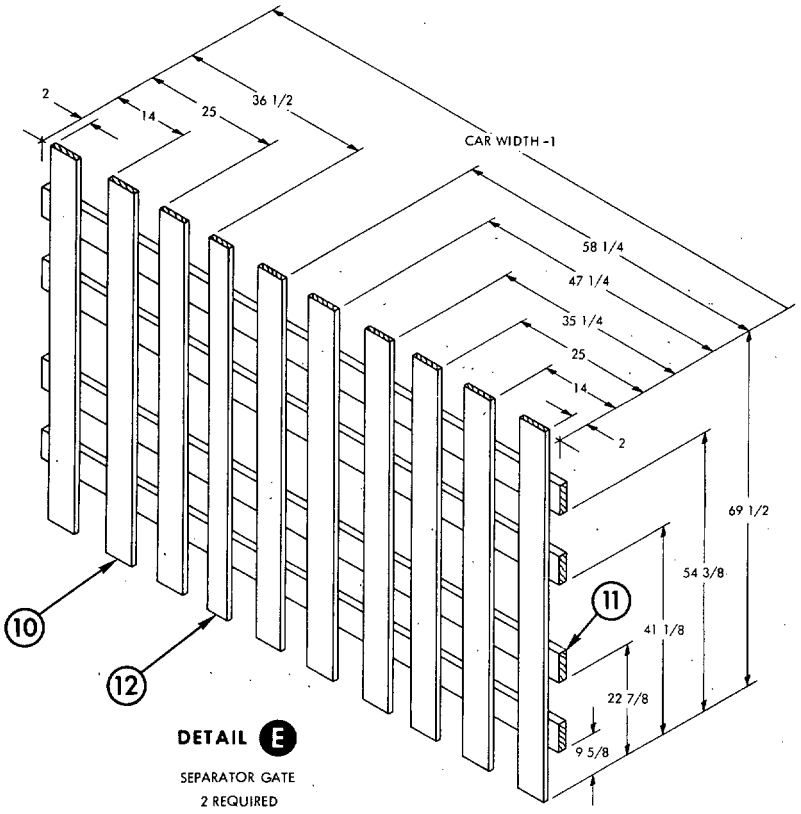
SEPARATOR GATE

6 REQUIRED

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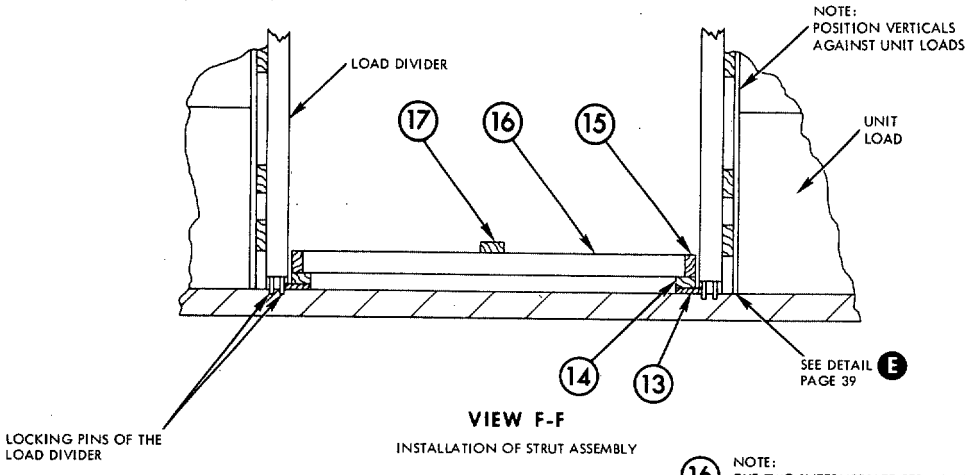
SECTION D-D



DETAIL E

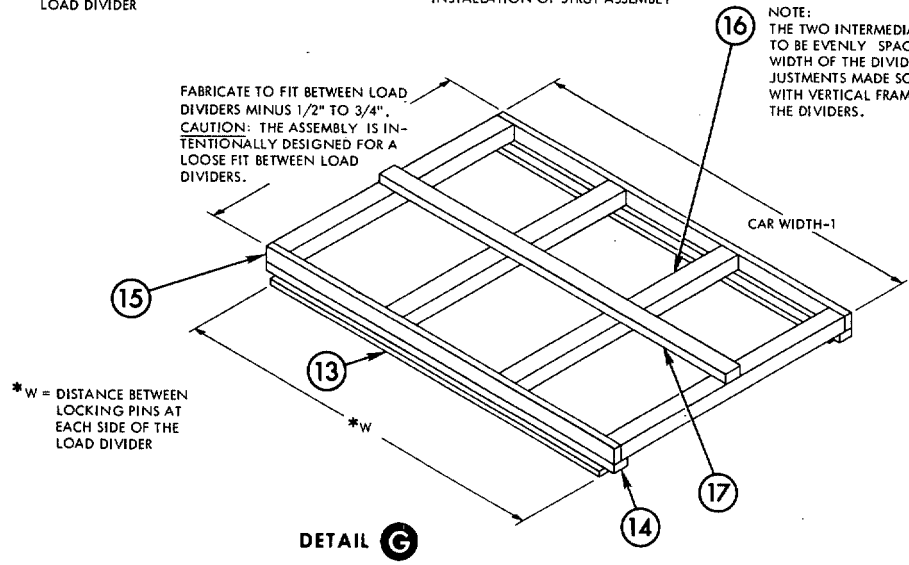
SEPARATOR GATE
2 REQUIRED

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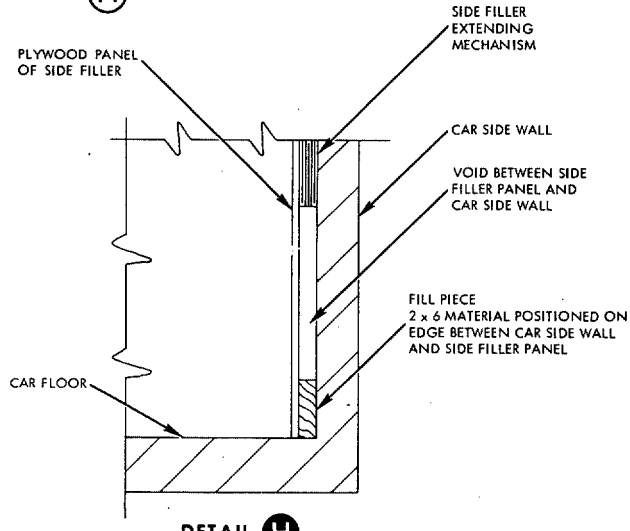
VIEW F-F

INSTALLATION OF STRUT ASSEMBLY



DETAIL G

STRUT ASSEMBLY
1 REQUIRED



DETAIL H

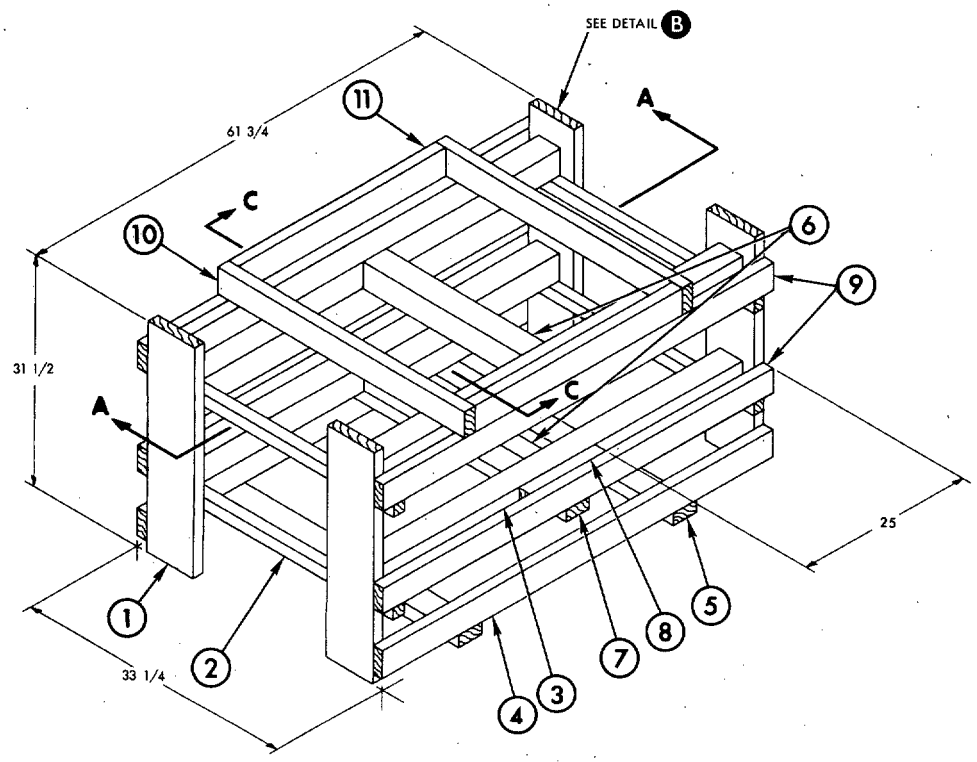
SHOWING REINFORCEMENT FOR
ADJUSTABLE SIDE FILLERS

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DUMMY UNIT LOAD



NOTE: PIECES 10 & 11 ARE ONLY REQUIRED WHEN OVERSTOWING ANOTHER DUMMY UNIT LOAD

NOTES:

(1) TOENAIL STRUT TO VERTICAL WITH 16d NAILS, 3 EACH END (TOP AND BOTH SIDES) ALSO END NAIL THROUGH VERTICAL WITH 16d NAILS, 2 EACH END.

(2) TOENAIL CROSS BRACE TO STRUT WITH 16d NAILS, 3 EACH END (TOP AND BOTH SIDES)

(3) NAIL LATERAL MEMBER TO PIECES 3, 8, AND 9 WITH 16d NAILS, 1 PER JOINT, AND TO CROSS BRACE (PIECE 6) WITH 4-16d NAILS.

(4) NAIL BUFFER STRIP TO VERTICAL WITH 16d NAILS, 2 PER JOINT, AND TO PIECE 8 WITH 4-16d NAILS.

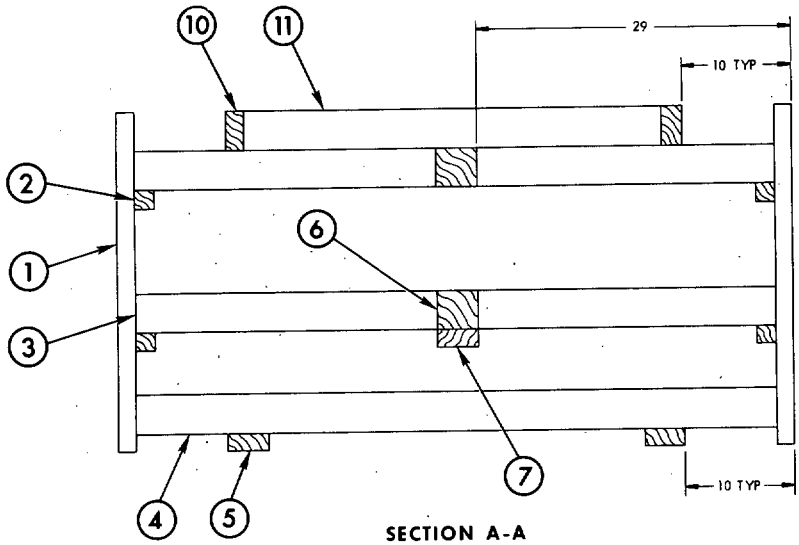
(5) NAIL UPPER SUPPORT PIECE (PIECE 10) TO PIECE 11 WITH 16d NAILS, 2 PER JOINT, TOENAIL TO STRUT (PIECE 3) WITH 10d NAILS, 2 PER JOINT (1 EACH SIDE).

(6) TOENAIL UPPER SUPPORT PIECE (PIECE 11) TO PIECE 9 WITH 10d NAILS, 1 PER FOOT.

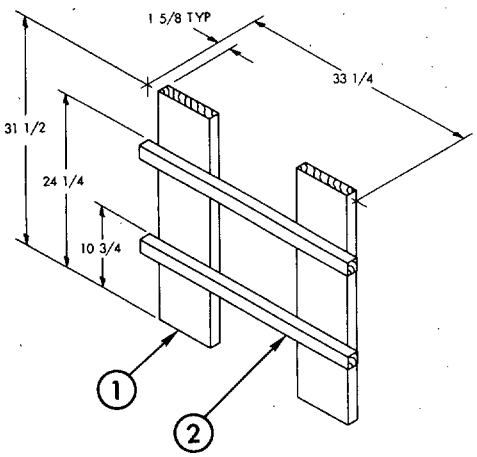
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NAILS	
					NUMBER	SIZE
11	UPPER SUPPORT PIECE	2 x 4 x 38 1/2	2	9	SEE NOTE (6)	
10	UPPER SUPPORT PIECE	2 x 4 x 33 1/4	2	3, 11	SEE NOTE (5)	
9	BUFFER STRIP	2 x 4 x 61 3/4	4	1, 8	SEE NOTE (4)	
8	SPACER	2 x 4 x 12	4	3	4	10d
7	LATERAL MEMBER	2 x 4 x 33 1/4	1	3, 6, 8, 9	SEE NOTE (3)	
6	CROSS BRACE	4 x 4 x CUT TO FIT	2	3	SEE NOTE (2)	
5	SUPPORT PIECE	2 x 4 x 33 1/4	2	4	2 PER JOINT	16d
4	LONGITUDINAL	2 x 4 x 61 3/4	2	1	2 PER JOINT	16d
3	STRUT	4 x 4 x 58 1/2	4	1	SEE NOTE (1)	
2	STRUT CLEAT	2 x 2 x 33 1/4	4	1	3 PER JOINT	16d CLNHD
1	VERTICAL	2 x 8 x 31 1/2	4	SEE 2	-	-

LIST OF MATERIALS & NAILING DATA

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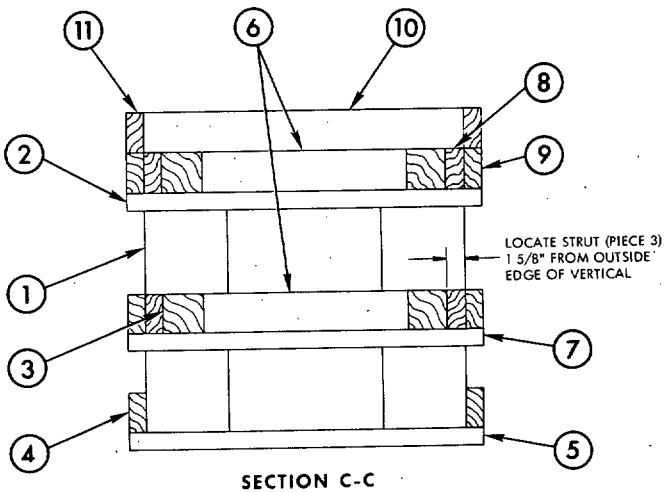


SECTION A-A



DETAIL B

END FRAME
2 REQUIRED



SECTION C-C

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REVIEW ACTIVITY:
NAVY-OS, AS

PREPARING ACTIVITY:
NAVY-OS
(PROJECT NO. 8140-N212)

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